Week 12 · Four Tilting Thrusters, and the Open-Loop Check

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-11-03

ℹ Recordings of this semester's own classes

Every session of this course is recorded as it is given and posted to the playlist below. The written material here is the reference; the recording is the lecture, at the pace it was actually delivered. Use it to review a derivation, or to catch up on a missed week before the laboratory.

	Where
[충남대] 2026 센서신호처리 — this course, recorded live, 2026	playlist

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	**Control Engineering** (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	**Control System Design** (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	**Advanced Control Engineering** (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	**Sensor Signal Processing and Fusion** (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	**Capstone Design** (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

✎ Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	`winget install --id Git.Git -e`
once	the folder that will hold the course, e.g. `Documents`	`git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git`
before every class	inside the cloned folder `Sensor-Signal-Processing-and-Fusion`	`git pull`

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder, and `_tools\mss_path.m` finds it there in either release layout — `Tools\MSS\VESSELS\otter.m` in the older releases, or `Tools\MSS\CRAFT\USV\models\otter.m` in the current master.
- **The two releases are not the same vessel.** The figures printed in these notes were measured with the older layout. Terminal surge speed agrees at 1.0286 m/s, but the current master turns 25.5 % faster under differential thrust, 0.1276 rad/s against 0.1017 rad/s. `mss_path` therefore prefers the older layout when both are present, and warns when only the newer one is available, so a reading that cannot reproduce is announced rather than silent.
- **A 118 MB folder disappears on the first pull after 2026-10-27.** `lectures\MSS-master` held a copy of MSS until then. It sat outside `Tools\MSS`, so no course script could reach it, and it only made cloning slower. Its removal is expected and is not a fault. Any file kept inside that folder must be moved elsewhere before pulling.

- **Course**: USV Guidance, Navigation and Control (Graduate)
- **Department**: Autonomous Vehicle System Engineering, Chungnam National University
- **This week**: ① a tilting thruster occupies **two** columns of B, not one ② the four places `otter.m` must change to carry four of them ③ eight numbers given by hand, and the three asymmetries that make the symmetric guess wrong

★ Before starting

- Week 3's column rule must be available from memory: a thrust applied at a point contributes one column to B, and the rule that builds that column is the only definition of B this course uses.
- Week 6's change to the motion model must be understood as a change to the **model**, not an addition to its input. `otter_bow.m` carries the bow thrust inside τ; this week does the same thing for four tilting units.
- Verify both in one command: `cd lectures/W06_simulink; W06_0_setup` must print the three-thruster configuration without error, and `verify_otter_bow` must print `PASS`.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. State the column rule for a thrust applied at (x_i, y_i), and show from it that a fixed thruster occupies one column of B while a tilting thruster occupies two.
2. Write the generalised force of four tilting thrusters as $\boldsymbol{\tau} = B\mathbf{f}$ with $B \in \mathbb{R}^{3\times 8}$, state its rank and the dimension of its null space, and confirm that $\delta_i = 0$ recovers the two-thruster expression of `otter.m` exactly.
3. Identify the four places in `otter.m` that must change to carry four tilting units, and state why the hull hydrodynamics are not among them.
4. Predict the surge force left over when a shaft speed $+n$ is opposed by $-n$, compute the astern speed that cancels it, and confirm both against the measured table.
5. Derive $N = \bar{x}\,Y$ with $\bar{x} = (l_f - l_a)/2$ for four equally tilted thrusters, predict the residual yaw moment of a symmetric sway command to three decimal places, and derive the thrust ratio that removes it.
6. Distinguish a check that the equations are right from a check that the model is wired to those equations, and state why re-integrating the logged inputs outside Simulink is the second of these while a single-instant acceleration comparison is neither.
7. State what this week deliberately leaves out, and why the three measured asymmetries make an allocation necessary rather than optional.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b or later, with Simulink
MSS toolbox	reachable at `Tools\MSS` ; `_tools\mss_path.m` adds it
Weeks assumed	Week 3 (the column rule and B), Week 5 (saturation and limits), Week 6 (changing the motion model itself)
Plant	`_tools\otter_tilt4.m` — `otter.m` with four tilting units, same hull
Configuration	`otter_config('quad_tilt')`
Model	`W12_openloop.slx` , written by `W12_1_build_openloop.m`
Solver	fixed-step `ode4` , $h = 0.02$ s
Scope of this week	**open loop only.** No allocation, no controller, no guidance

Part 1 · Principle and experiment, section by section

Every section states what is observed, says what follows from it, and then runs **the experiment that measures it**, on its own model. The numbers quoted in the text are printed by the run a few lines below it.

Part of a section	What it holds
What is observed	the effect itself, with the numbers it produces
What follows from it	numbered lines, one step each
Experiment 12-x	the model, the commands, the output actually obtained, the figure, and a table that reads every feature of the figure back to **the numbered line that predicts it**

12-0. Setting up (10 min)

```
cd lectures/W12_simulink
W12_1_build_openloop          % once — writes W12_openloop.slx and img/*.png
W12_0_setup                   % every session — defines every constant
```

`W12_1_build_openloop` prints one line, and both counts must be zero:

```
  W12_openloop   overlaps 0   diagonals 0
```

`W12_0_setup` prints the configuration this week works in:

```
  Week 12  four tilting thrusters, open loop
    thruster stations   x = [+1.50 +1.50 -1.00 -1.00] m
                        y = [-0.395 +0.395 -0.395 +0.395] m
    servo limit         +45 deg,  rate 90 deg/s
    shaft limits        -101.737 .. +103.931 rad/s
    thrust at +-60      +39.888 N ahead, -23.202 N astern   (k_neg/k_pos = 0.5817)
    cancelling astern    -78.670 rad/s, not -60
    B                   3 x 8, rank 3, null space 5
    command             n = [60 60 60 60] rad/s
                    delta = [45 45 45 45] deg
```

Four lines of that block are the week in miniature. The shaft limits are not symmetric, the two thrust coefficients are not equal, B has five more columns than rows, and the astern speed that cancels $+60$ rad/s is not -60.

> **⚠ The shaft command is in rad/s, not rpm**
>
> `n_cmd = [60 60 60 60]` is 60 radians per second, about 573 rpm. `otter.m` has always used rad/s and `otter_tilt4.m` keeps that, because the thrust coefficients k_{pos} and k_{neg} are quoted against it. Entering 600 expecting rpm asks for 5730 rpm, and the model saturates it to 103.931 rad/s.

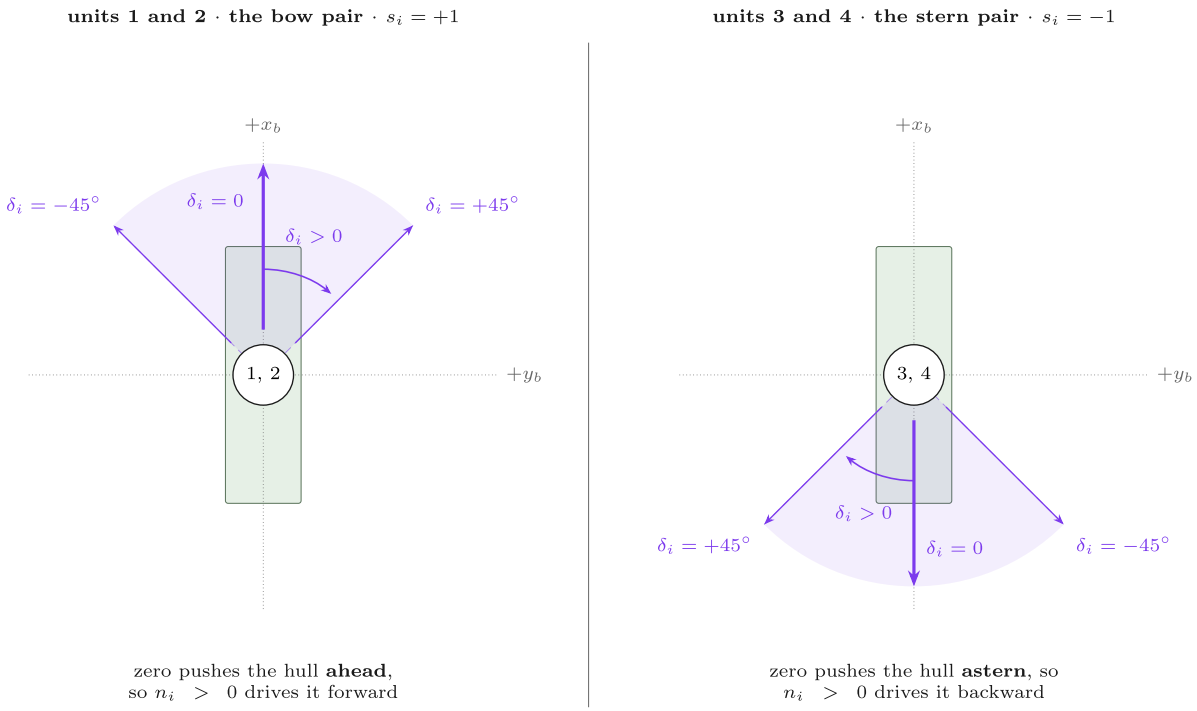
12-1. Four thrusters, two columns each, and two different zeros

This section answers: what does a tilting thruster change about the allocation matrix, before any allocation is attempted?

What is observed. Experiment 12-1 reports that the forward pair at $\delta = 0$ reproduces `otter.m` exactly, that the aft mounting sign flips $\boldsymbol{\tau}$ to 3.55×10^{-15}, that $\boldsymbol{\tau}$ agrees with $B\mathbf{f}$ to 7.19×10^{-14} over 200 random states, and that B is 3×8 with rank 3.

- four thrust stations: two at $x_b = +1.50$ m, two at $x_b = -1.00$ m, each 0.395 m off centreline
- each unit turns through $\pm 45°$ at up to $90°$/s
- the two pairs are mounted facing **opposite** ways, so $\delta_i = 0$ drives the hull ahead at the bow and astern at the stern
- eight numbers are commanded; three components of force result

The mounting comes first, because every column of B carries it.



The bow is up in both cells. Each pair measures δ_i from its *own* zero towards y_b, so a positive δ_i turns clockwise here in both. Only the zero differs, and that difference is the sign s_i carried by the columns of B.

The result. Write $s_i = +1$ when unit i's zero points ahead and $s_i = -1$ when it points astern. The thrust direction and the force of unit i are then

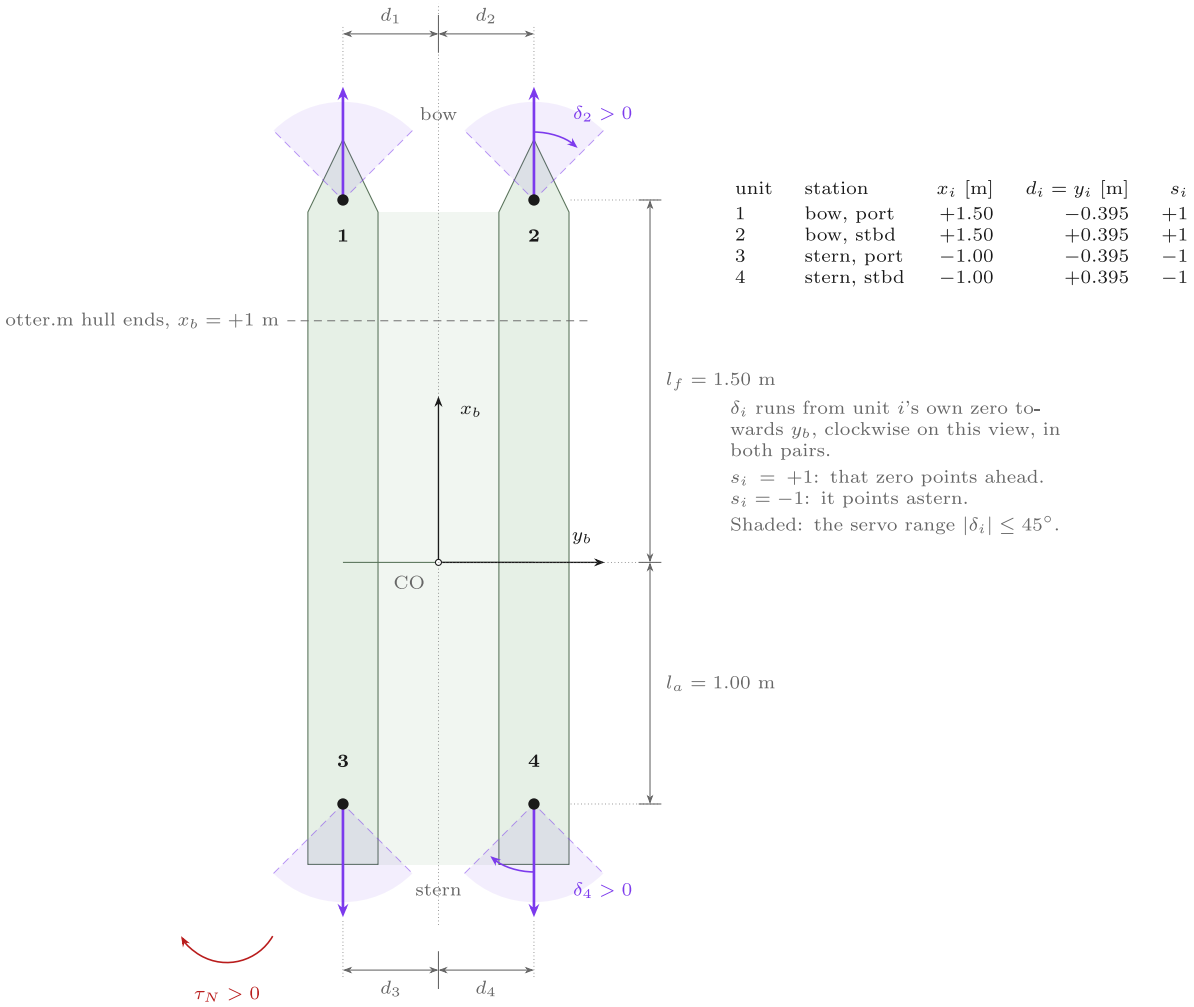
$$\mathbf{e}_i = s_i \begin{bmatrix} \cos\delta_i \\ \sin\delta_i \end{bmatrix}, \qquad \mathbf{f}_i = T_i \begin{bmatrix} \cos\delta_i \\ \sin\delta_i \end{bmatrix}, \qquad s = [+1 \quad +1 \quad -1 \quad -1]$$

The sign is deliberately kept out of $\mathbf{f}$ and placed in B, so that $\mathbf{f}$ holds only what the hardware reports. With the column rule applied to each component separately, the force on the hull is linear in those eight numbers:

$$\boldsymbol{\tau} = \begin{bmatrix} X \\ Y \\ N \end{bmatrix} = B\mathbf{f}, \qquad \mathbf{f} = \begin{bmatrix} T_1\cos\delta_1 \\ T_1\sin\delta_1 \\ \vdots \\ T_4\cos\delta_4 \\ T_4\sin\delta_4 \end{bmatrix}, \qquad \mathbf{b}_i^c = s_i \begin{bmatrix} 1 \\ 0 \\ -y_i \end{bmatrix}, \quad \mathbf{b}_i^s = s_i \begin{bmatrix} 0 \\ 1 \\ x_i \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 & 0 & -1 \\ 0.395 & 1.500 & -0.395 & 1.500 & -0.395 & 1.000 & 0.395 & 1.000 \end{bmatrix}$$

Symbol	Quantity	Value · source
x_i	longitudinal station of unit i	$+1.50, +1.50, -1.00, -1.00$ m — `otter_config.m`, field `pos`
$y_i = d_i$	transverse station of unit i	$-0.395, +0.395, -0.395, +0.395$ m — same
s_i	mounting sign: which way the zero points	$+1, +1, -1, -1$ — `otter_config.m`, field `mount`
δ_i	tilt angle, from unit i's own zero towards y_b	commanded, clipped to $\pm 45°$
T_i	thrust magnitude of unit i	$k_{pos} n_i \lvert n_i \rvert$ for $n_i \geq 0$, $k_{neg} n_i \lvert n_i \rvert$ otherwise
$\mathbf{b}_i^c, \mathbf{b}_i^s$	the two columns unit i owns	the $\cos$ column and the $\sin$ column
B	allocation matrix	3×8, rank 3 — built by `_tools/otter_B.m` from the column rule
l_f, l_a	forward and aft moment arms	1.50 m, 1.00 m — `otter_config.m`, fields `l_f`, `l_a`



Derivation, one line at a time.

1. The force–moment triple produced by a planar force $\mathbf{g} = [g_x \;\; g_y]^\mathsf{T}$ applied at (x, y) in body axes is $[g_x,\; g_y,\; x g_y - y g_x]^\mathsf{T}$. This is Week 3's column rule, and it is the only definition of B this course uses.
2. For a **fixed** thruster, $\mathbf{e}_i$ is a constant of the hull. The single unknown is T_i, the rule returns one column, and four fixed thrusters give $B \in \mathbb{R}^{3\times 4}$.
3. For a **tilting** thruster, $\mathbf{e}_i = \mathbf{e}_i(\delta_i)$. The map $(T_i, \delta_i) \mapsto T_i \mathbf{e}_i$ is polar-to-Cartesian, invertible wherever $T_i > 0$, so the unit carries two independent numbers rather than one.
4. Applying line 1 to the two components of $s_i T_i [\cos\delta_i \;\; \sin\delta_i]^\mathsf{T}$ separately gives the columns $s_i[1, 0, -y_i]^\mathsf{T}$ and $s_i[0, 1, x_i]^\mathsf{T}$. Four units give $B \in \mathbb{R}^{3\times 8}$, printed numerically above.

5. B has rank 3: rows 1 and 2 are independent by inspection, and row 3 is not a combination of them because the x_i and y_i are not all zero. Its null space has dimension $8 - 3 = 5$.
6. Five null-space dimensions are five independent ways to change the eight numbers without changing the force. Choosing among them is allocation — the minimum-norm choice is $\mathbf{f} = B^{\dagger}\boldsymbol{\tau}$ with $B^{\dagger} = B^{\mathsf{T}}(BB^{\mathsf{T}})^{-1}$ — and it is next stage's subject, not this one's.
7. Multiplying out $B\mathbf{f}$ gives the scalar form the motion model evaluates:

$$X = \sum_{i=1}^{4} s_i T_i \cos\delta_i, \qquad Y = \sum_{i=1}^{4} s_i T_i \sin\delta_i, \qquad N = \sum_{i=1}^{4} s_i\left(x_i T_i \sin\delta_i - y_i T_i \cos\delta_i\right)$$

8. Two consequences follow at $\delta = 0$, and Experiment 12-1 measures both. Every sine dies, so $Y = 0$ and $N = -\sum s_i y_i T_i$: the differential-thrust yaw of `otter.m`, with four stations in place of two. And because $s_3 = s_4 = -1$, the aft pair alone reproduces `otter.m` with $\boldsymbol{\tau}$ reversed.

Experiment 12-1 · The new model contains the old one (15 min)

What it measures. Lines 1, 7 and 8 of the derivation: that the column rule and the scalar form agree, and that $\delta = 0$ returns `otter.m` — forward with $\boldsymbol{\tau}$ as it stands, aft with $\boldsymbol{\tau}$ reversed.

The model. None. `verify_otter_tilt4` calls `otter_tilt4` and `otter` directly, over 200 random states, so the check is on the equations and not on any wiring.

Opening and running.

```
verify_otter_tilt4            % 네 가지 항등식 / four identities
```

Expected output:

```
verify_otter_tilt4
  stations   x = [+1.50 +1.50 -1.00 -1.00] m
             y = [-0.395 +0.395 -0.395 +0.395] m
  tilt limit +-45 deg each
  B          3 x 8, rank 3, null space 5

(a) aft mounting sign flips tau exactly :  3.55e-15
(b) fore pair, delta 0 == otter.m       :  0.00e+00
(c) tau                == otter_B * f   :  7.19e-14
(d) saturation clamps n AND delta       :  0.00e+00

thrust at +-60 rad/s   +39.888 N ahead, -23.202 N astern  (k_neg/k_pos = 0.5817)
cancelling astern speed   -78.670 rad/s   (not -60)
limits                   -101.737 .. +103.931 rad/s

PASS  (tol 1e-09, 200 random states)
```

Reading the output against the derivation.

Where to look	What is there	Which line predicts it
`(b)`	exactly zero, not merely small	line 8 — at $\delta = 0$ the forward pair evaluates the arithmetic of `otter.m`, in the same order
`(a)`	3.55×10^{-15}, one rounding step	line 8 — $s_3 = s_4 = -1$, so the aft pair gives `otter.m` with every component of $\boldsymbol{\tau}$ reversed
`(c)`	7.19×10^{-14} over 200 random states	line 7 — the scalar form and $B\mathbf{f}$ are one expression in two floating-point orders
`(d)`	zero	the model clips **both** n and δ, so a command outside the envelope is received at the envelope
`rank 3, null space 5`	printed, not asserted	lines 5 and 6

What the output says

- The four-unit model is a strict extension: nothing in it changes what the first eleven weeks measured.

What to try	What to watch
`verify_otter_tilt4` twice in a row	only `(c)` changes, in its last digits — the random states feed the $B\mathbf{f}$ comparison alone

> ✎ **In class**
>
> - **Purpose** — fix two ideas: a tilting unit is two unknowns, and the mounting sign lives in B.
> - **Point to** — the third row of B: each pair contributes $s_i(-y_i)$ then $s_i(+x_i)$, and that is the whole of the column rule.
> - **Ask** — "Eight unknowns and three equations. How many solutions?" Five dimensions of them, which is why an allocator is a choice and not a calculation.
> - **Take away** — the forward pair at $\delta = 0$ **is** `otter`, to the last bit; the aft pair is `otter` running backwards.

12-2. What had to change inside the motion model

This section answers: which lines of `otter.m` carry the thruster configuration, and which lines carry the hull?

What is observed. `_tools/otter_tilt4.m` is MSS `otter.m` copied and changed in four places, each marked in the file with `% ---- CHANGED (i of 4)`. The four marks sit in the same four places as Week 6's `otter_bow.m`.

- the rigid-body and added-mass matrices, the Coriolis terms, the restoring forces, the linear and cross-flow damping, the nonlinear yaw damping $N_h = N_r(1 + 10|r|)r$ and the kinematics are character-for-character identical
- the hull is the same hull, and the file has to show that

The result. Four changes, and nothing else.

	`otter.m`	`otter_tilt4.m`	**Why**
1 · input check	`length(n) ~= 2`	`length(n) ~= 4` and `length(delta) ~= 4`, plus an eighth argument `quad`	the configuration arrives as data, so the stations and tilt limits are not hard-coded in the plant
2 · thruster data	l_1, l_2	`px`, `py`, `d_max`, `smount` read out of `quad`, and B built from them	with four stations a pair of signed arms is not enough, and the mounting sign has to reach the columns
3 · thrust loop	`for i = 1:1:2`, n saturated	`for i = 1:1:4`, n **and** δ saturated	derivation line 3 below — the servo is part of the vessel, not of the controller
4 · the force vector	`tau = [T1+T2 0 0 0 0 -l1*T1-l2*T2]'`	`tau3 = B_thr * f`	derivation lines 4 to 6 — one matrix product in place of a literal

Derivation, one line at a time.

1. The propellers are unchanged. $k_{pos} = 0.02216/2$ and $k_{neg} = 0.01289/2$ stay at the values `otter.m` sets, because the same propeller is fitted four times rather than twice.
2. The shaft limits are therefore also unchanged: $n_{max} = +103.931$ rad/s and $n_{min} = -101.737$ rad/s, derived in `otter.m` from the maximum thrust each way.
3. The tilt servo belongs inside the plant. A servo that refuses $70°$ refuses it whatever asked, so the clip must be in the motion model; a clip placed only in the controller would let a mis-wired command produce a force the vessel cannot make.
4. `otter.m` writes its $\boldsymbol{\tau}$ as a literal six-row vector because two fixed thrusters reach only two of the six rows. Here every unit reaches rows 1, 2 and 6 at once, so the literal is replaced by the product $\boldsymbol{\tau}_3 = B\mathbf{f}$ of §12-1.
5. The split is deliberate. B is built once, from geometry and mounting alone, and holds every sign; $\mathbf{f}$ is rebuilt each step and holds only T_i and δ_i. Nothing in the force code can disagree with the figure, because the figure is B.
6. Rows 3, 4 and 5 stay zero. No thruster on this vessel acts in heave, roll or pitch, and the trim moment that does act in pitch is applied elsewhere in the file, exactly as in `otter.m`.

The second change, which is the column rule and the mounting sign written as code:

```
% ---- CHANGED (2 of 4) ----------------------------------------------------
px     = quad(1,:);                    % x stations (m)
py     = quad(2,:);                    % y stations (m)
d_max  = quad(3,:);                    % tilt limits, both ways (rad)
smount = quad(4,:);                    % mounting signs, +1 forward, -1 aft

% 유효행렬을 열 규칙(3주차)으로 **여기서** 만든다. 이름이 B_thr 인 것은
% otter.m 의 B 가 이미 **선폭**이라서다 — 그 줄은 고치지 않는다.
% The effectiveness matrix, built here from the Week 3 column rule. It is named
% B_thr because B is already the beam in otter.m, and that line is not touched.
B_thr = zeros(3,8);
for i = 1:1:4
    for k = 1:1:2                      % k = 1 gives e = s(1,0), k = 2 e = s(0,1)
        e    = zeros(2,1);
        e(k) = smount(i);
        B_thr(:, 2*i-2+k) = [ e(1)
                              e(2)
                              px(i)*e(2) - py(i)*e(1) ];
    end
end
% --------------------------------------------------------------------------
```

The fourth change, which is the matrix product itself:

```
% ---- CHANGED (4 of 4) ----------------------------------------------------
% Control forces and moments — written as a matrix product
% otter.m:  tau = [T1+T2  0  0 0 0  -l1*T1 - l2*T2]'      <- sway is zero
% here:     tau_3 = B f.  Each tilting thruster contributes TWO columns of B
%           and two entries of f, so the 3-by-8 B is constant and the whole
%           dependence on the commands sits in f.
%
%     f = [ T_1 cos d_1 ; T_1 sin d_1 ; ... ; T_4 cos d_4 ; T_4 sin d_4 ]
%
% 쌍마다 0 도가 다른 것은 B 안에 있다 — otter_B 가 거치 부호 s = [1 1 -1 -1]
% 을 열에 실어 두었다. 그래서 여기에는 부호가 하나도 없다.
% The per-pair zero lives inside B: otter_B has already carried the mounting
% sign s = [1 1 -1 -1] into the columns, so no sign appears here.
f = reshape([Thrust(:) .* cos(delta(:)), ...
             Thrust(:) .* sin(delta(:))].', 8, 1);

tau3 = B_thr * f;                      % [X ; Y ; N]

tau = [tau3(1)
       tau3(2)
       0
       0
       0
       tau3(3)];
% --------------------------------------------------------------------------
```

In the code	In the derivation	Note
`e(k) = smount(i)`	$\mathbf{e}_i = s_i[\cos\delta_i \quad \sin\delta_i]^{\mathsf{T}}$	the only place the mounting sign is applied
`px(i)*e(2) - py(i)*e(1)`	$x_i e_{iy} - y_i e_{ix}$	§12-1 line 1 — the column rule, third entry
`B_thr(:, 2*i-2+k)`	the two columns unit i owns	$k=1$ is $\mathbf{b}_i^c$, $k=2$ is $\mathbf{b}_i^s$
the `reshape(...,8,1)`	$\mathbf{f}$ of §12-1	the cosine and the sine of each unit, interleaved in that order
`tau3 = B_thr * f`	$\boldsymbol{\tau} = B\mathbf{f}$	no sign and no station appears here: both are already in B
rows 3, 4, 5 left at zero	line 6	heave, roll and pitch carry no thruster

Change 3 is the loop that fills `Thrust` and clips both commands. It keeps `otter.m`'s shaft-speed clip and its k_{pos} / k_{neg} branch unaltered, and adds the servo clip `delta(i) > d_max(i)` — which §12-4 measures as the third asymmetry.

> **The sway row is no longer structurally zero**
>
> Week 6 made the same point about `otter_bow.m`. A motion model whose τ_y is identically zero describes an under-actuated vessel, however a sway force is added to it afterwards. The four-unit model earns a non-zero second row from the geometry itself, which is why the eight numbers of §12-3 can be given directly to the plant with no allocator between.

12-3. The force eight numbers make

This section answers: given eight numbers by hand, is the resulting force the one the column rule predicts?

What is observed. Experiment 12-3 commands eight sets of eight numbers and compares two independent computations of $\boldsymbol{\tau}$: the product $B\mathbf{f}$ of §12-1, and the force measured back out of `otter_tilt4` from the acceleration it produces. The worst disagreement over the eight scenarios is 2.8×10^{-14} N or N·m.

- the eight scenarios isolate one effect each, and are reused by §12-4, §12-5 and §12-6
- S8 asks for $70°$, and the table records what the model received

The result. The model is driven open loop: eight constants on the canvas, a saturation stage, the plant, and the log.

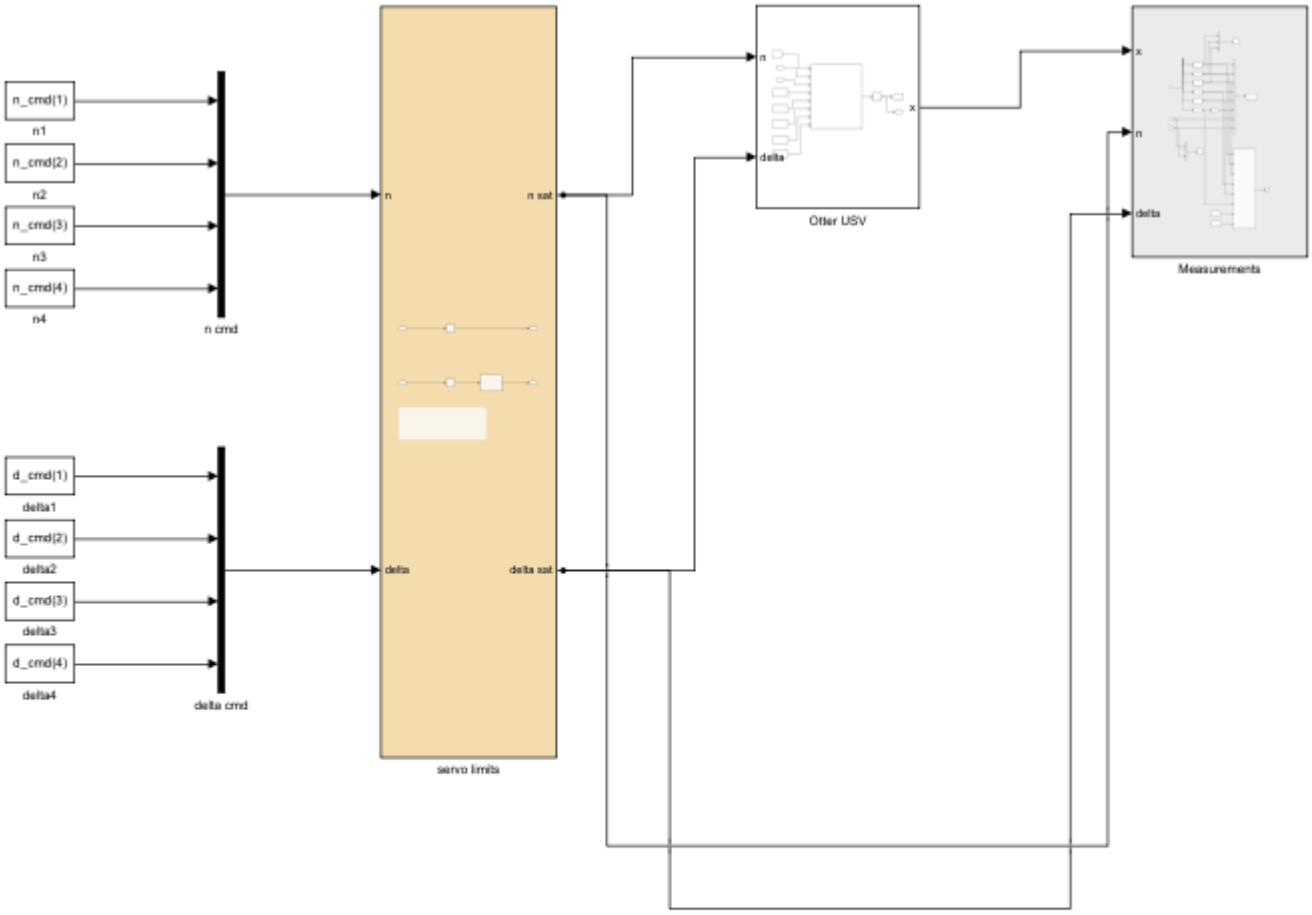
$$\boldsymbol{\tau}_{\text{measured}} \;=\; M\left(\dot{\mathbf{x}}_{\text{on}} - \dot{\mathbf{x}}_{\text{off}}\right)_{1:6}$$

Symbol	Quantity	Value · source
M	6×6 mass matrix, rigid body plus added mass	from `otter_tilt4` at the same state
$\dot{\mathbf{x}}_{\text{on}}$	acceleration with the commanded (n, δ)	`otter_tilt4(x, n, delta, …)`
$\dot{\mathbf{x}}_{\text{off}}$	acceleration with $n = 0$	the same call with the thrust removed
residual	$\max\lvert B\mathbf{f} - [X\ Y\ N]^{\mathsf{T}}\rvert$	asserted below 10^{-12} inside the script

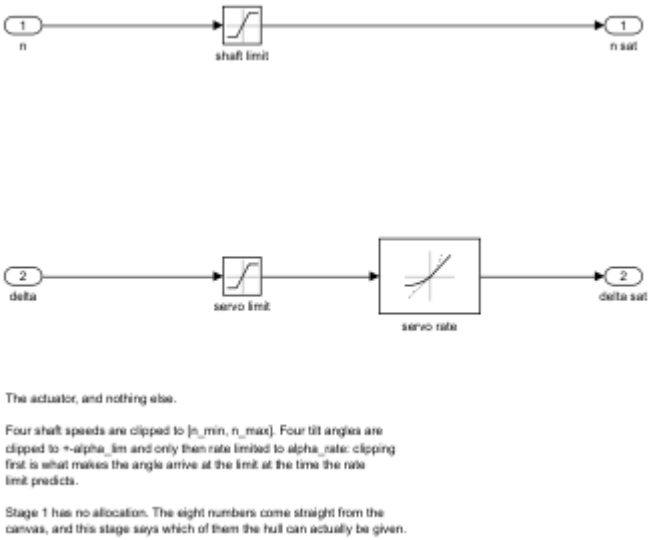
Derivation, one line at a time.

1. The six-row acceleration of the vessel is $\dot{\boldsymbol{\nu}} = M^{-1}(\boldsymbol{\tau} - \mathbf{d}(\boldsymbol{\nu}, \mathbf{x}))$, where $\mathbf{d}$ collects damping, Coriolis and restoring terms. Those terms depend on the state but not on the thrust.
2. Evaluating the same state twice, once with the thrust and once with $n = 0$, therefore differences $\mathbf{d}$ away: $\dot{\boldsymbol{\nu}}_{\text{on}} - \dot{\boldsymbol{\nu}}_{\text{off}} = M^{-1}\boldsymbol{\tau}$.

3. Multiplying back by M recovers $\boldsymbol{\tau}$ exactly, not approximately, because acceleration is affine in the thrust. This is the same argument Week 6 used to show its superposition had been exact.
4. The comparison is run **at rest**, $\mathbf{x} = \mathbf{0}$, so that no state-dependent term can hide a sign error by cancelling it.
5. The tilt command is clipped before it reaches the column rule, so the table prints the clipped angle. A case that asks for more than $45°$ must therefore produce the row of the case that asked for exactly $45°$.



Eight Constant blocks carry the eight numbers, two Mux bars gather them into vectors, the tan subsystem applies the physical envelope, the green block is `otter_tilt4`, and the grey block logs `[u v r N E psi]` followed by the post-servo (n, δ) tapped from the branches below.



The upper lane clips the shaft speed to $[-101.737, +103.931]$ rad/s. The lower lane clips the angle to $\pm 45°$ and only then rate-limits it to $90°$/s. The two lanes never meet, because shaft speed and tilt angle are independent commands.

> **★ The angle is clipped before it is rate-limited, not after**
>
> Rate-limiting first would let the limiter chase a $70°$ target through the legal range and sit at $45°$ only on arrival, so the recorded angle would depend on how long the run lasted. Clipping first makes the target legal, and the rate limiter then describes only how fast the servo gets there.

Experiment 12-3 · Eight commands, and the force each one makes (20 min)

What it measures. Lines 3 and 5: that the force measured out of the plant is the force the column rule predicts, and that the servo clip is visible in the table.

The model. `W12_openloop.slx`. Between runs only the eight constants change; the wiring does not.

Opening and running. The runner takes a selection, so one scenario can be re-examined without redrawing the other seven.

```
W12_0_setup
open_system('W12_openloop')    % Run — 궤적과 상태량이 뜬다 / the track appears
W12_V_openloop                 % 여덟 전부 / all eight, and the three tables
W12_V_openloop('S5')           % 하나만 / one of them, its own sheet redrawn
W12_V_openloop({'S5','S6'})    % 둘을 견준다 / two, to be compared
```

Expected output, the first of three tables printed by that script:

```
(1) hand formula against the motion model, at rest

    tau from the hand formula    X = sum si Ti cos di
                                 Y = sum si Ti sin di
                                 N = sum si (xi sin di - yi cos di) Ti
                                 s = [+1 +1 -1 -1], the mounting signs
    tau measured out of otter_tilt4's acceleration, same state

  case  n [rad/s] after clip       delta [deg] clipped       X [N]    Y [N]  N [N m]    resid
  ------------------------------------------------------------------------------------------
  S1    [ 60.0  60.0 -60.0 -60.0]  [   0    0    0    0]   126.180    0.000    0.000  0.0e+00
  S2    [-60.0 -60.0  60.0  60.0]  [   0    0    0    0]  -126.180   -0.000    0.000  1.4e-14
  S3    [ 60.0  60.0  60.0  60.0]  [  45   45   45   45]     0.000    0.000  141.025  2.8e-14
  S4    [ 60.0  60.0  60.0  60.0]  [ -45  -45  -45  -45]     0.000    0.000 -141.025  2.8e-14
  S5    [ 60.0  60.0  60.0  60.0]  [  45   45  -45  -45]     0.000  112.820   28.205  1.4e-14
  S6    [ 60.0  60.0  73.5  73.5]  [  45   45  -45  -45]   -28.205  141.025    0.000  2.2e-14
  S7    [ 60.0  60.0 -60.0 -60.0]  [  45   45    0    0]   102.814   56.410   84.615  2.8e-14
  S8    [ 60.0  60.0  60.0  60.0]  [  45   45  -45  -45]     0.000  112.820   28.205  1.4e-14
  ------------------------------------------------------------------------------------------
  worst residual 2.8e-14 N or N m over the 8 scenarios
```

Reading the table against the derivation.

Where to look	What is there	Which line predicts it
the `resid` column	at most 2.8×10^{-14}	line 3 — the two computations are the same expression, so only rounding separates them
S1, $X = 126.180$ N	$2(39.888 + 23.202)$	§12-1 line 8 at $\delta = 0$: the bow pair ahead, and the stern pair ahead too, because $s_i = -1$ reverses a negative shaft speed
S2, $X = -126.180$ N	the same magnitude, mirrored	the mounting sign swaps which pair runs astern, so the asymmetric thrust curve cancels out of the total
S3, $X = Y = 0$ with $N = 141.025$ N·m	$2(l_f + l_a)T \sin 45°$	§12-1 line 7 — the four force vectors sum to zero and only their moments survive
S5, $N = 28.205$ N·m although a pure sway was asked	$2(l_f - l_a)T \sin 45°$	§12-4 — the stations are not symmetric about CO
S6, the commanded 73.5 rad/s	$60\sqrt{1.5}$	§12-4 — the stern speed that cancels that moment, at the price of -28.205 N of surge
S8 identical to S5	the `delta` column reads 45, although 70 was asked	line 5, and §12-2

What the table says

- The eight numbers reach the hull as the column rule says they will, and nothing in the wiring alters them.

What to try	What to watch
`W12_V_openloop('S1')`	the S1 sheet alone: surge only, and the vessel of Weeks 1 to 11 is back
`W12_V_openloop({'S5','S6'})`	two sheets and a two-curve overlay: the leftover moment, and the same crab with that moment cancelled

> ✎ **In class**
>
> - **Purpose** — separate "the equations are right" from "the vessel does what was wanted".
> - **Point to** — the S1 and S2 rows. Both reach 126.180 N although the thrust curve is far from symmetric, because the mounting sign decides which pair runs astern.
> - **Ask** — "S8 asked for $70°$. Which column shows that it did not get it?" The `delta` column, which prints what the plant received.
> - **Take away** — a residual of 10^{-14} says the arithmetic is right; it says nothing about whether the force was the right force.

12-4. Three asymmetries that break the symmetric guess

This section answers: why does a command that looks symmetric produce a motion that is not?

What is observed. Three asymmetries run through the eight rows of table (1), and each one is a property of the vessel rather than of the command.

- the thrust curve: $+60$ rad/s gives $+39.888$ N, -60 rad/s gives only -23.202 N
- the stations: $l_f = 1.50$ m forward against $l_a = 1.00$ m aft
- the servo: $70°$ asked, $45°$ given

The result. Each has a closed form, and each predicts a number in the table.

$$\underbrace{n_{rev} = -n\sqrt{\frac{k_{pos}}{k_{neg}}} = -78.670\ \text{rad/s},}_{\text{thrust curve}} \qquad \underbrace{N = \bar{x}\,Y, \quad \bar{x} = \frac{l_f - l_a}{2} = 0.25\ \text{m}}_{\text{stations}}$$

Symbol	Quantity	Value · source
k_{pos}	thrust coefficient, ahead	$0.02216/2 = 0.011080$ — `otter.m`, unchanged
k_{neg}	thrust coefficient, astern	$0.01289/2 = 0.006445$ — same
k_{neg}/k_{pos}	the asymmetry ratio	0.5817 — printed by `W12_0_setup`
n_{rev}	astern speed matching $+60$ rad/s	-78.670 rad/s — printed by `verify_otter_tilt4`
$\bar{x}$	longitudinal centroid of the four stations	0.25 m
N	residual yaw of a four-unit symmetric crab	$+28.205$ N·m — table (1), row S5

Derivation, one line at a time.

1. The thrust of one propeller is the signed square $T = kn|n|$, with $k = k_{pos}$ for $n \geq 0$ and $k = k_{neg}$ for $n < 0$. Both coefficients are positive, and $k_{neg} < k_{pos}$ because a propeller designed to push ahead is less efficient in reverse.
2. Running one unit at $+n$ and another at $-n$ therefore does not cancel: $T(+n) + T(-n) = (k_{pos} - k_{neg})n^2$.
3. Row S1 runs the bow pair at $+60$ rad/s and the stern pair at -60. Each stern unit returns -23.202 N, which its mounting sign $s_i = -1$ turns into $+23.202$ N of ahead force, so the four push ahead in unequal shares — 39.888 N forward against 23.202 N aft, a ratio of 1.72. Row S2 swaps which pair runs astern, so $|X|$ is the same

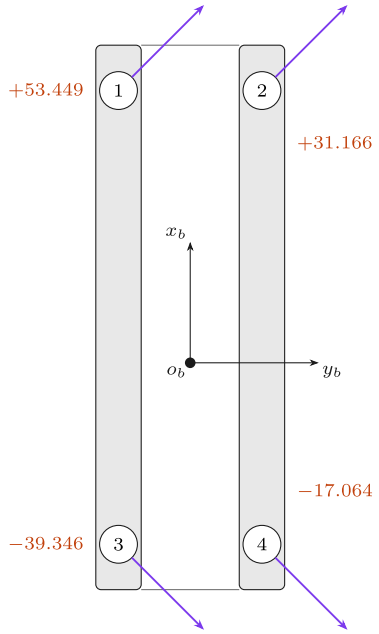
126.180 N in both. The thrust curve does not break the fore–aft symmetry of the total; it breaks **the split between the pairs**, and that split decides each pair's moment as soon as the units tilt.

4. Setting $k_{pos}n^2 + k_{neg}n_{rev}|n_{rev}| = 0$ with $n_{rev} < 0$ gives the astern speed at which a unit matches a forward one: $n_{rev} = -n\sqrt{k_{pos}/k_{neg}} = -60 \times 1.3112 = -78.670$ rad/s. `verify_otter_tilt4` prints it as `cancelling astern speed`.
5. The second asymmetry needs no thrust curve at all. Take row S5 — the bow pair at $\delta = +45°$, the stern pair at $-45°$, all four at the same thrust T. With the mounting signs every unit makes the **same** sway force $f_y = T\sin 45° = 28.205$ N, while the surge forces alternate: $+f_x$ forward, $-f_x$ aft. Then

$$N = \sum_i (x_i f_y - y_i f_{ix}) = f_y \sum_i x_i - \sum_i y_i f_{ix}$$

6. The second sum is zero: inside each pair the two units sit at $\pm y$ with the same f_{ix}, so their two contributions cancel. The first sum is not zero: $\sum x_i = 2l_f - 2l_a = 1.0$ m, so $N = 2(l_f - l_a)f_y$.
7. Dividing by $Y = 4f_y$ puts it in the form that explains itself: $N = \bar{x}Y$ with $\bar{x} = \frac{1}{4}\sum x_i = (l_f - l_a)/2 = 0.25$ m. Four equal sway forces are one sway force acting at the centroid of the four stations, and that centroid is 0.25 m ahead of o_b. With $Y = 112.820$ N this is $N = 112.820 \times 0.25 = 28.205$ N·m, the S5 and S8 reading.

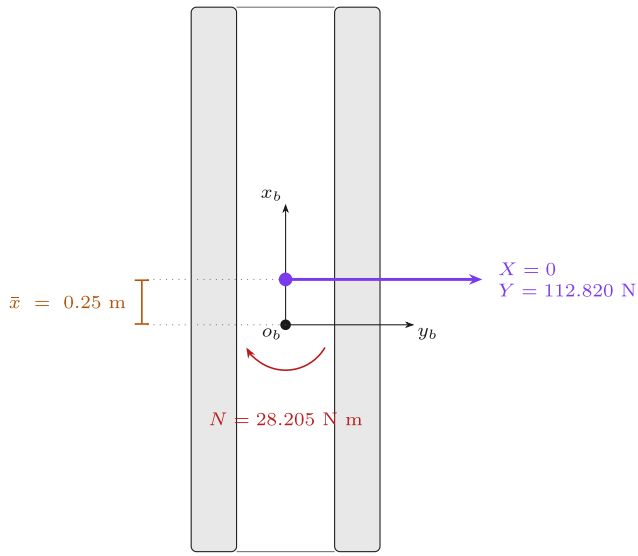
(a) four equal sway forces, four unequal arms



Every unit is given $n_i = 60$ rad/s, the bow pair at $\delta_i = +45°$ and the stern pair at $-45°$. With s_i the four *sway* forces come out equal, $f_{iy} = 28.205$ N, and the surge forces cancel fore against aft.

$x_i f_{iy} - y_i f_{ix}$ [N m] is printed beside each unit. The f_{ix} terms cancel inside each pair, at $\pm y$, and what is left is $N = f_y \sum x_i = 2\,(l_f - l_a)\,f_y$

(b) the same four forces, moved to one point



Four equal sway forces are one sway force at the centroid of the four stations:

$$\bar{x} = \tfrac{1}{4}\textstyle\sum x_i = \dfrac{l_f - l_a}{2} = 0.25 \text{ m}$$

That point is not o_b, so a sway command given symmetrically to four thrusters pulls 0.25 m ahead of the origin and yaws: $N = Y\,\bar{x} = 112.820 \times 0.25 = 28.205$ N m

8. The condition that removes it follows from the same sum with the bow pair at T_f and the stern pair at T_a, all at $|\delta| = 45°$: $\sum x_i T_i = 3.0T_f - 2.0T_a = 0$, hence $T_a = 1.5T_f$. With $T_f = 39.888$ N this needs $T_a = 59.832$ N, that is $n_a = \sqrt{59.832/k_{pos}} = 60\sqrt{1.5} = 73.485$ rad/s. Row S6 runs it and reads $N = 0.000$ N·m — at the price of $X = 2\cos 45°\,(T_f - T_a) = -28.205$ N. The moment did not vanish; it became a surge of the same number.
9. The third asymmetry is the servo, and it is not a sign or a ratio but a wall. δ is clipped before the column rule sees it, so the force is computed from $45°$ whatever was asked. A command of $70°$ and a command of $45°$ are the same command to this vessel.

10. All three together are the argument for an allocator. The eight numbers a person would guess from symmetry are not the eight numbers that produce the wanted force, and solving for them by hand — as line 8 did, for one condition — does not scale to three conditions and a changing state.

> **⚠ No equal-thrust sway inside $\pm 45°$ is free of yaw**
>
> A symmetric crab leaves 28.205 N·m, and no common shaft speed removes it, because the asymmetry is in the geometry, not in the command. Line 8 buys the moment back with a surge of exactly -28.205 N. Unequal thrusts can do better — Assignment requirement 2 asks for such a set — but four equal ones never can.

Experiment 12-4 · The three asymmetries, measured (20 min)

What it measures. Lines 3, 4, 7, 8 and 9, each against a row of table (1).

The model. The same `W12_openloop.slx`. `W12_tab_track` prints the three asymmetries below table (3), from the same eight runs.

Opening and running.

```
W12_V_openloop                % 여덟 줄 전부 / all eight rows
W12_V_openloop({'S5','S6'})   % 두 번째 비대칭만 / the second asymmetry alone
```

Expected output, the block printed below table (3):

```
      the three asymmetries this week turns on
        thrust curve   +60 rad/s gives +39.888 N, -60 rad/s gives -23.202 N, so
                       the pair running forward carries 1.72 times the pair
                       running astern -- and yet S1 and S2 reach the SAME
                       +126.180 N, because the mirrored mounting swaps which
                       pair is the one running astern
        the stations   l_f = 1.50 m forward, l_a = 1.00 m aft, so S5 crabs with
                        +28.205 N m of yaw left over, and S6 cancels that moment
                       at n = 73.485 rad/s only by letting  -28.205 N of surge
                       appear: no EQUAL-thrust sway inside +-45 deg is yaw-free
        the servo      S8 asks for 70 deg and is given 45 deg, so its row of
                       table (1) is S5's row exactly
```

Reading the output against the derivation.

Where to look	What is there	Which line predicts it
`1.72 times`	the $39.888/23.202$ split inside S1	line 3 — k_{pos}/k_{neg}
`the SAME +126.180 N`	the S1 and S2 X columns	line 3 — the mounting signs swap which pair runs astern
`+28.205 N m of yaw`	the S5 and S8 N columns	line 7 — $\bar{x}Y$ with $\bar{x} = 0.25$ m
`-28.205 N of surge`	the S6 X column, with $N = 0.000$	line 8 — the moment traded for a surge
`given 45 deg`	the S8 `delta` column	line 9

What to try	What to watch
`W12_V_openloop('S6')` alone	$N = 0.000$ N·m against S5's $+28.205$, and X going from 0 to -28.205 N
`d_deg = [90 90 -90 -90]'` in `W12_0_setup`, expecting a pure sway	the table returns S5's row. The servo refuses the extra $45°$ — line 9, not a sway

> ✎ **In class**
>
> - **Purpose** — show that the obstacle to commanding this vessel is the vessel, not the arithmetic.
> - **Point to** — cell (b) of the figure: one arrow, offset 0.25 m, and the yaw follows from the offset alone.
> - **Ask** — "Which of the three would disappear if the hull were symmetric fore and aft?" Only the second. The thrust curve and the servo belong to the hardware.
> - **Take away** — $\bar{x} = (l_f - l_a)/2 = 0.25$ m, and $N = \bar{x}Y$.

12-5. The track those forces produce

This section answers: does the vessel go where eight hand-chosen numbers send it?

What is observed. Experiment 12-5 runs each of the eight cases for 30 s from rest and reads four quantities at the end. S1 runs straight at 1.627 m/s; S5 crabs at $74.28°$ while turning; S8 lies exactly on S5.

- the readings are taken at $t = 30$ s, by when all eight have settled
- the crab angle is $\beta = \operatorname{atan2}(v, u)$ and the speed is $U = \sqrt{u^2 + v^2}$
- turn radius is $U/|r|$, reported as `straight` when $|r| < 10^{-4}$ rad/s

The result. A force held constant in body axes produces a steady body-frame velocity, and a steady velocity with a steady yaw rate is a circle.

$$U = \sqrt{u^2 + v^2}, \qquad \beta = \operatorname{atan2}(v, u), \qquad R = \frac{U}{|r|}$$

Symbol	Quantity	Value · source
U	total speed over the water	read at $t = 30$ s
β	crab angle, body x to the velocity	positive to starboard
R	radius of the steady turn	$U/\|r\|$; infinite when $r = 0$
T_{final}	run length for these readings	30 s, set inside `W12_tab_reint`

Derivation, one line at a time.

1. With (n, δ) held constant, $\boldsymbol{\tau}$ is constant in body axes, so the body-frame equation $M\dot{\boldsymbol{\nu}} + \mathbf{d}(\boldsymbol{\nu}) = \boldsymbol{\tau}$ is autonomous and settles on an equilibrium $\boldsymbol{\nu}^\star$ where damping balances thrust.
2. That equilibrium is reached in a few seconds for surge and sway, and in rather longer for yaw, because the nonlinear yaw damping $N_h = N_r(1 + 10|r|)r$ stiffens as r grows.
3. At equilibrium the body-frame velocity is fixed but the heading is not: $\dot{\psi} = r \neq 0$ rotates the velocity in the Earth frame, which traces a circle of radius $U/|r|$.
4. The crab angle β is a property of that equilibrium, not of the command. Row S7's force stands at $\operatorname{atan2}(56.410, 102.814) = 28.74°$ in body axes, yet the hull crabs at only $5.89°$, because sway damping on this hull is far larger than surge damping.
5. A case with $r = 0$ runs straight in the direction β, and three rows do: S1 at $\beta = 0$, S2 at $180°$, and S6 at $114.0°$ — a heading held while the hull travels sideways and slightly astern.
6. S8 must trace S5's circle exactly, since §12-4 line 9 showed the two commands reach the plant identically. The overlay draws S8 dashed so the coincidence is visible rather than hidden by overdrawing.

Experiment 12-5 · Eight tracks from eight commands (20 min)

What it measures. Lines 3 to 6: that each command settles, that the settled state is a circle or a straight line, and that S8 and S5 coincide.

The model. The same `W12_openloop.slx`, run eight times at $T_{final} = 30$ s. Each run gets **its own sheet** — `img/W12_result_S1.png` … `_S8.png` — because eight curves in one axes erase each other and the question of this section is whether each scenario did what was written down for it.

Opening and running.

```
W12_V_openloop                % 여덟 장 전부 / all eight sheets
W12_V_openloop('S5')          % 한 장만 다시 / that one sheet again
```

Expected output, the third table:

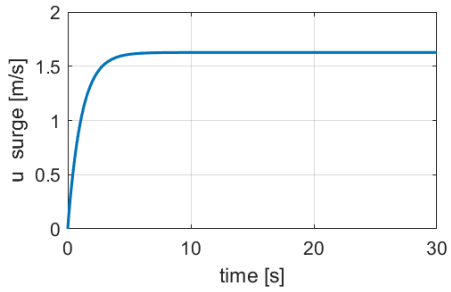
```
  (3) the motion those forces produce, scenario by scenario

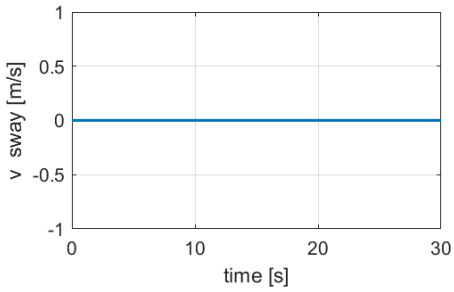
     every value is read at t = 30 s, by when all 8 have settled

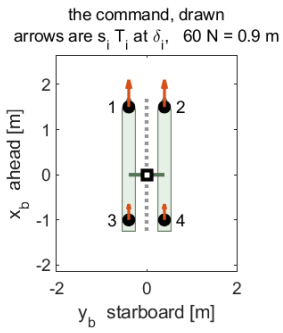
  case  command                      U [m/s]  crab [deg]  r [deg/s]  radius [m]
  ------------------------------------------------------------------------------
  S1    surge ahead                    1.627        0.00      0.000    straight
  S2    surge astern                   1.627      180.00      0.000    straight
  S3    yaw to starboard               0.030      -21.14     28.397        0.06
  S4    yaw to port                    0.030       21.14    -28.397        0.06
  S5    sway to starboard              0.747       74.28      7.589        5.64
  S6    sway, moment cancelled         0.895      113.99     -0.000    straight
  S7    turn while making way          1.464        5.89     20.239        4.15
  S8    tilt beyond the servo limit    0.747       74.28      7.589        5.64
  ------------------------------------------------------------------------------
```

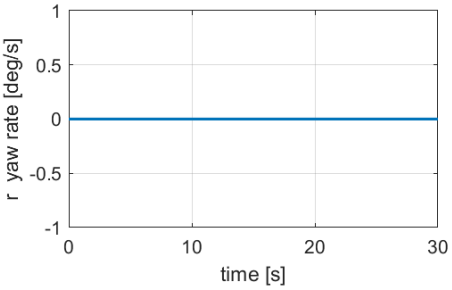
Each sheet carries the six states on the left, **the command drawn as four force arrows** at the top right, and the track with hull and heading below it. The arrow panel is what makes a wrong command distinguishable from a wrongly followed one.

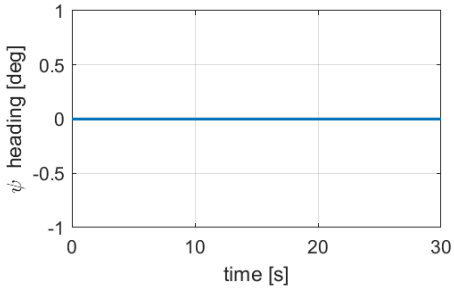
S1 surge ahead
n = [60.0 60.0 -60.0 -60.0] rad/s, δ = [0 0 0 0] deg, τ = [126.180 N, 0.000 N, 0.000 N m]
expected: runs straight ahead, no heading change: the hull settles at a steady u

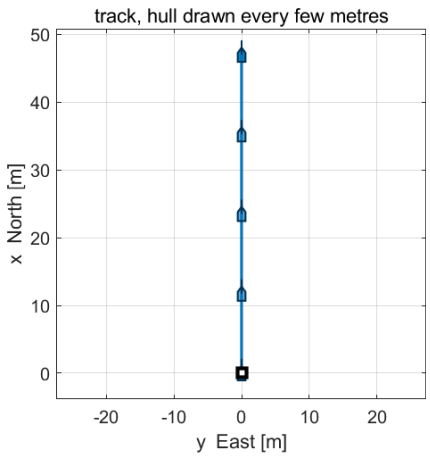


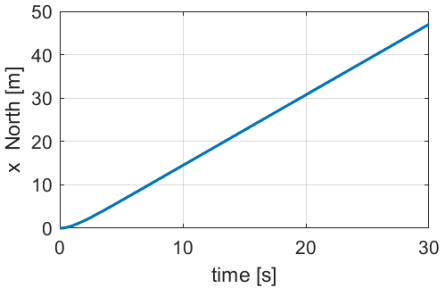


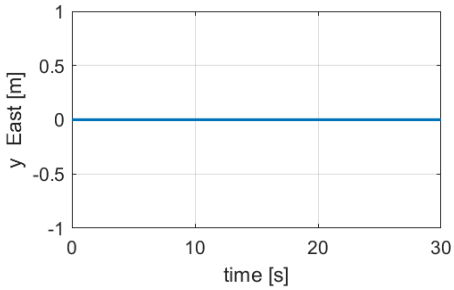




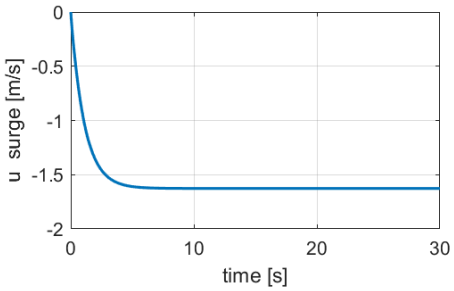


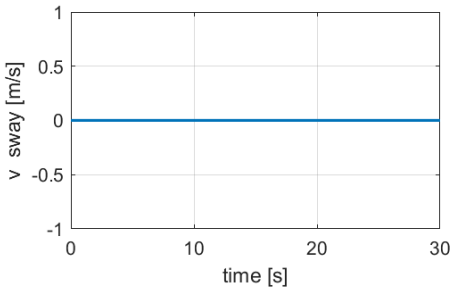


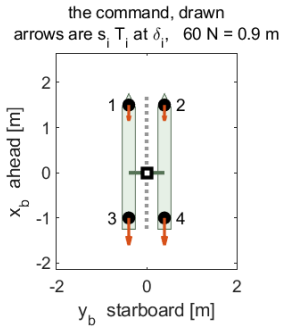


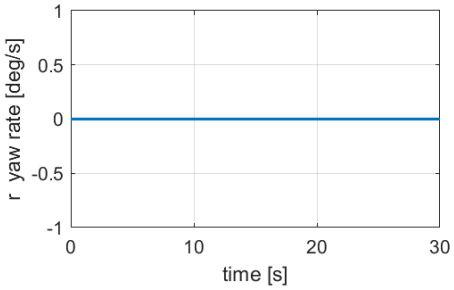


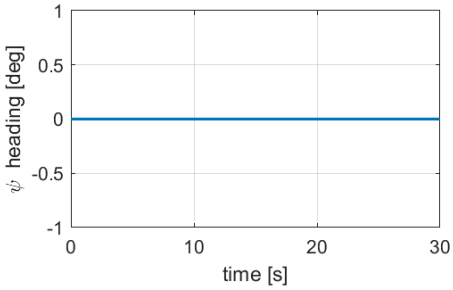
S2 surge astern
n = [-60.0 -60.0 60.0 60.0] rad/s, δ = [0 0 0 0] deg, τ = [-126.180 N, -0.000 N, 0.000 N m]
expected: runs straight astern with the SAME force magnitude as S1, mirrored

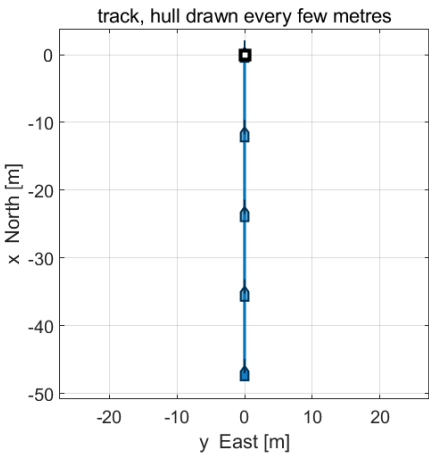


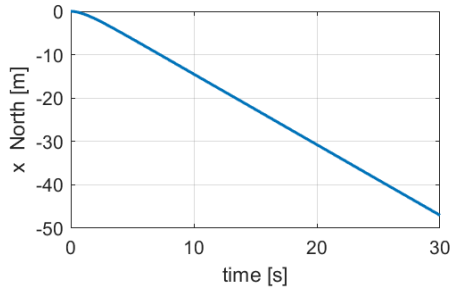


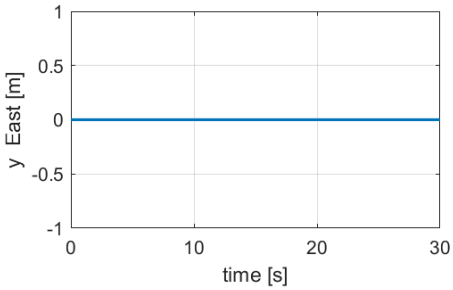




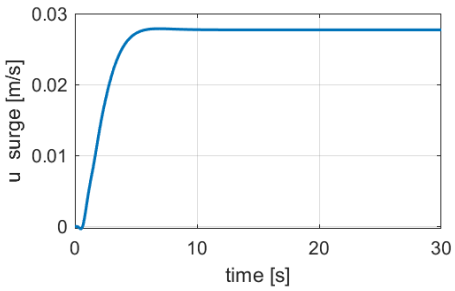


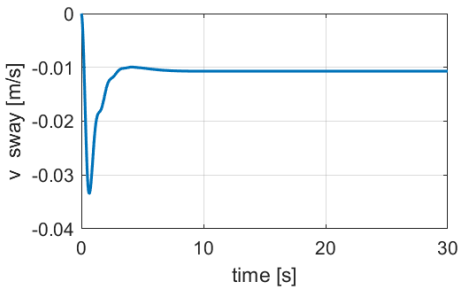


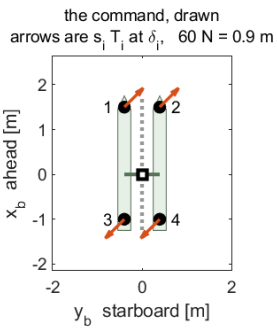


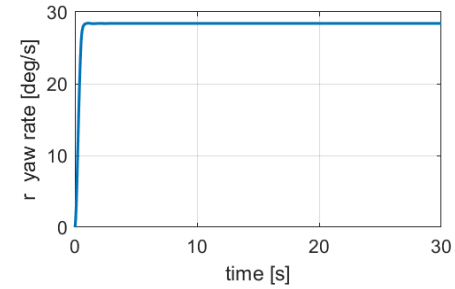


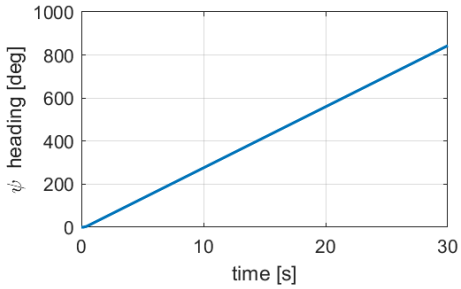
S3 yaw to starboard
n = [60.0 60.0 60.0 60.0] rad/s, δ = [45 45 45 45] deg, τ = [0.000 N, 0.000 N, 141.025 N m]
expected: turns on the spot to starboard: no net force at all, only a moment

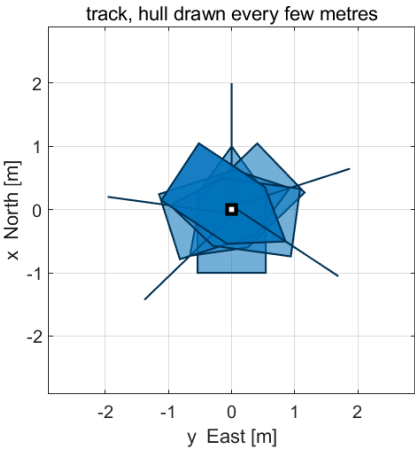


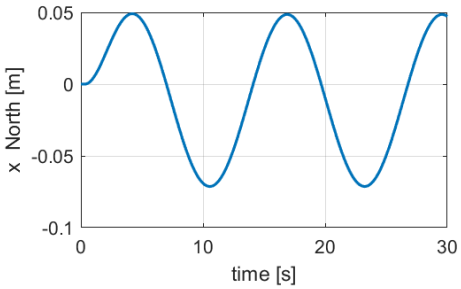


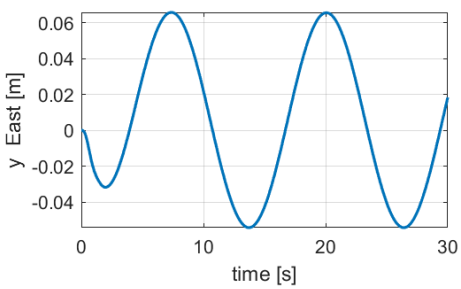




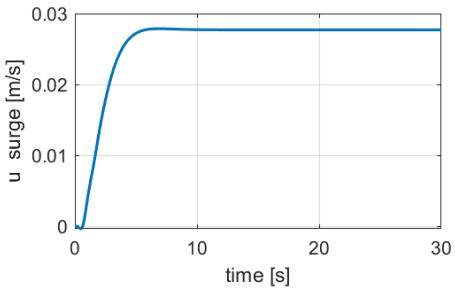


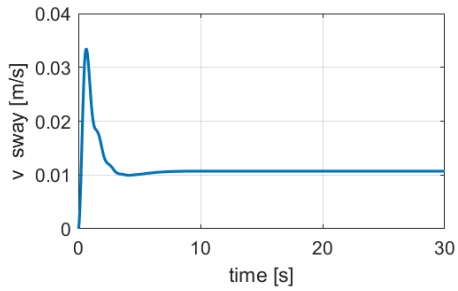


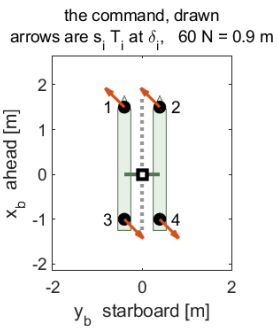


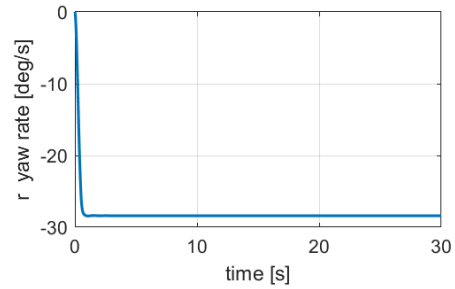


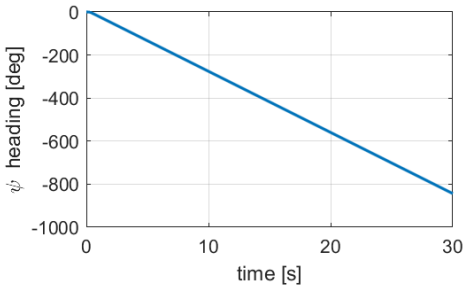
S4 yaw to port
n = [60.0 60.0 60.0 60.0] rad/s, δ = [-45 -45 -45 -45] deg, τ = [0.000 N, 0.000 N, -141.025 N m]
expected: turns on the spot to port, the exact mirror of S3

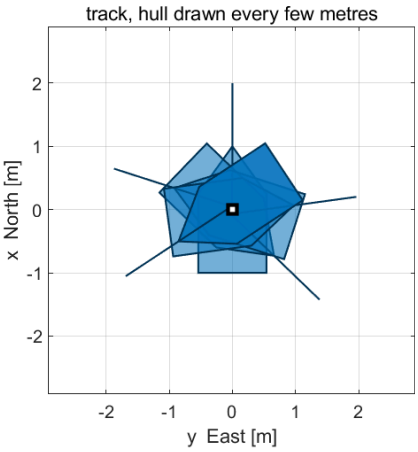


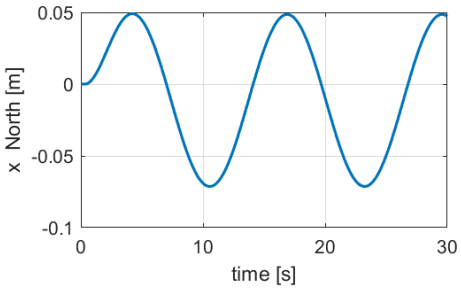


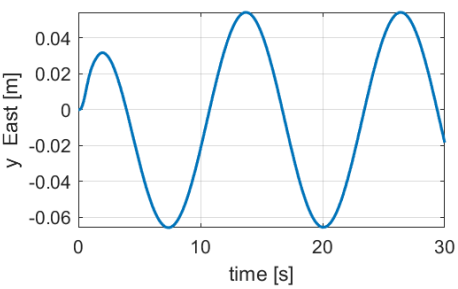




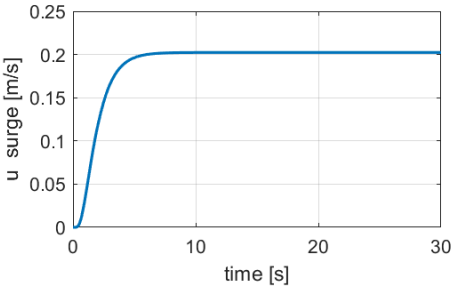


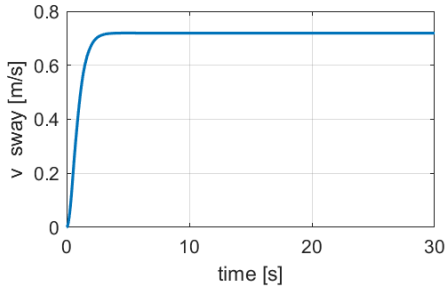


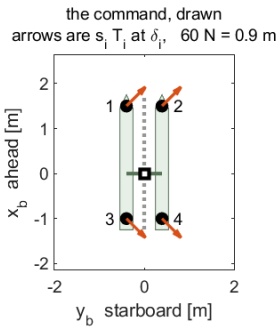


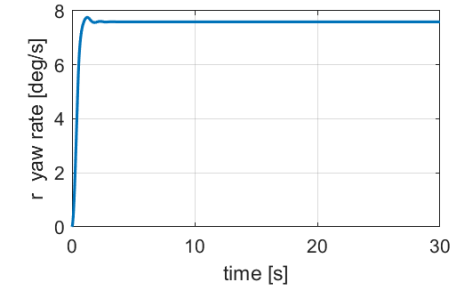


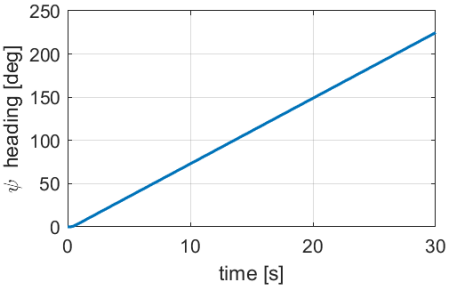
S5 sway to starboard
n = [60.0 60.0 60.0 60.0] rad/s, δ = [45 45 -45 -45] deg, τ = [0.000 N, 112.820 N, 28.205 N m]
expected: crabs to starboard with zero surge force, but keeps turning: $l_f > l_a$

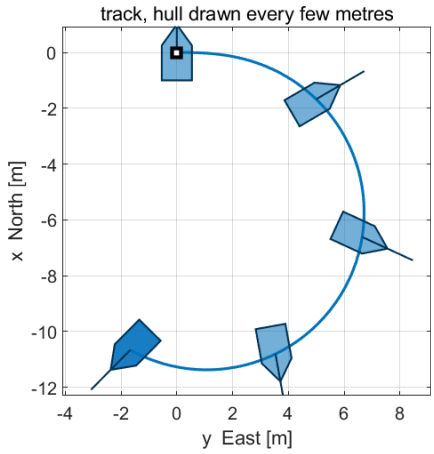


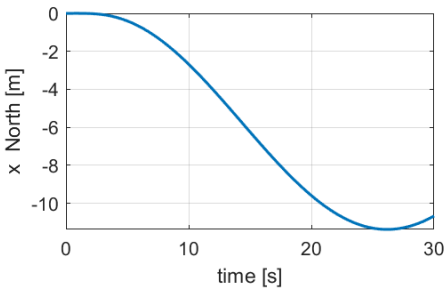


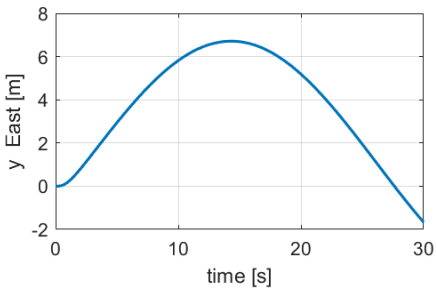




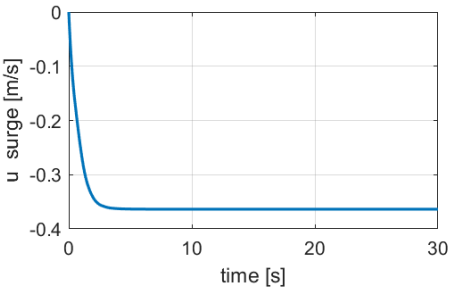


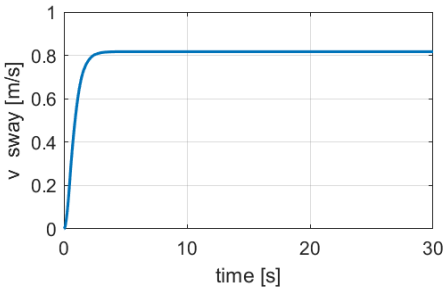


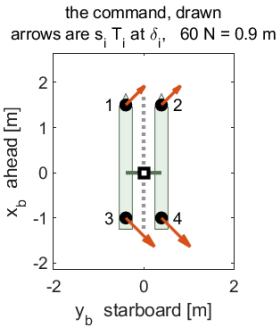


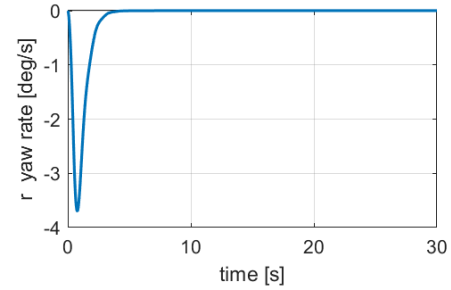


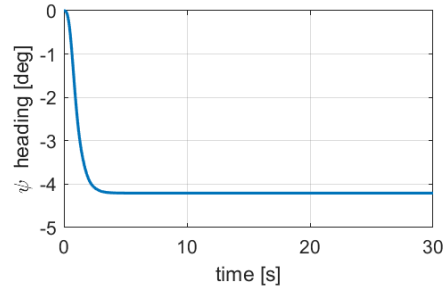
S6 sway, moment cancelled
n = [60.0 60.0 73.5 73.5] rad/s, δ = [45 45 -45 -45] deg, τ = [-28.205 N, 141.025 N, 0.000 N m]
expected: crabs to starboard on a steady heading, but now drifts ASTERN

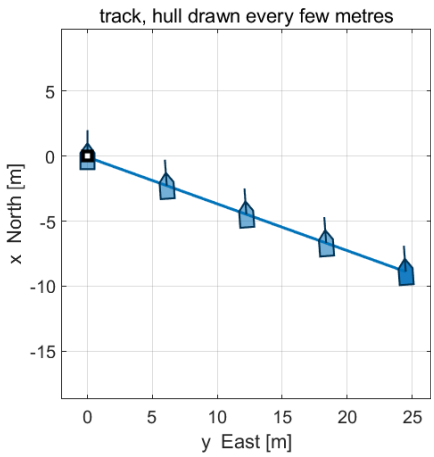


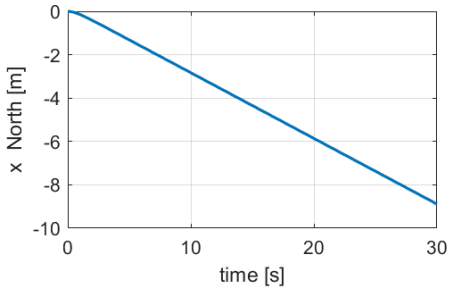


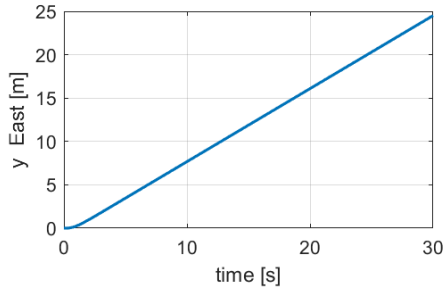




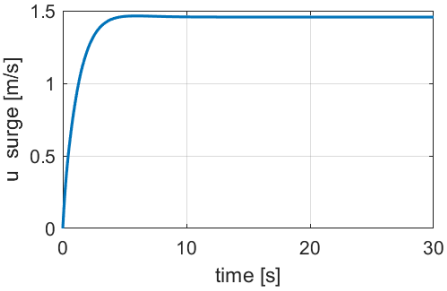


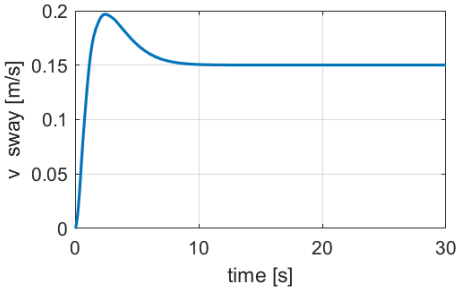


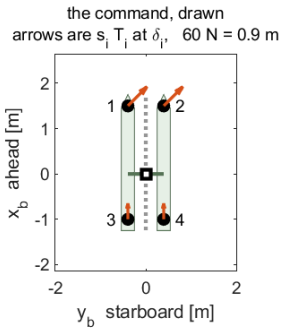


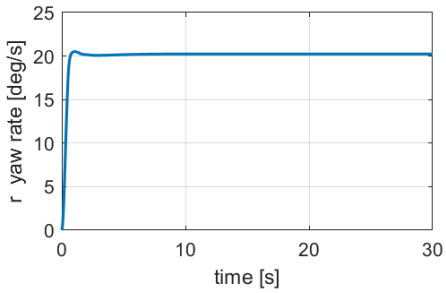


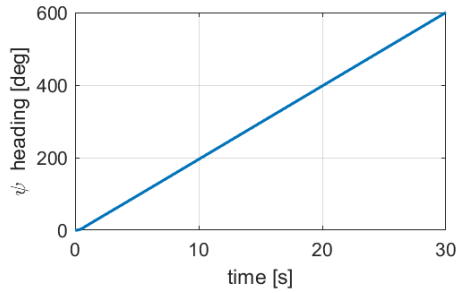
S7 turn while making way
n = [60.0 60.0 -60.0 -60.0] rad/s, δ = [45 45 0 0] deg, τ = [102.814 N, 56.410 N, 84.615 N m]
expected: surge, sway and yaw at once: a turning circle, the most ship-like row

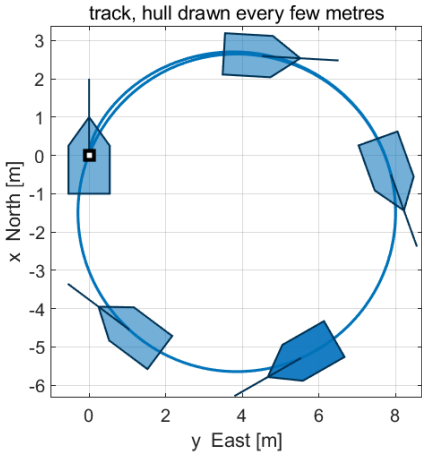


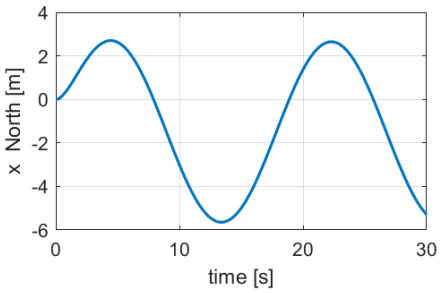


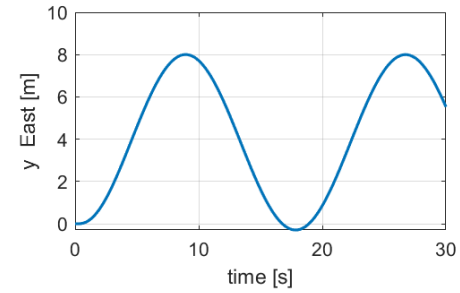




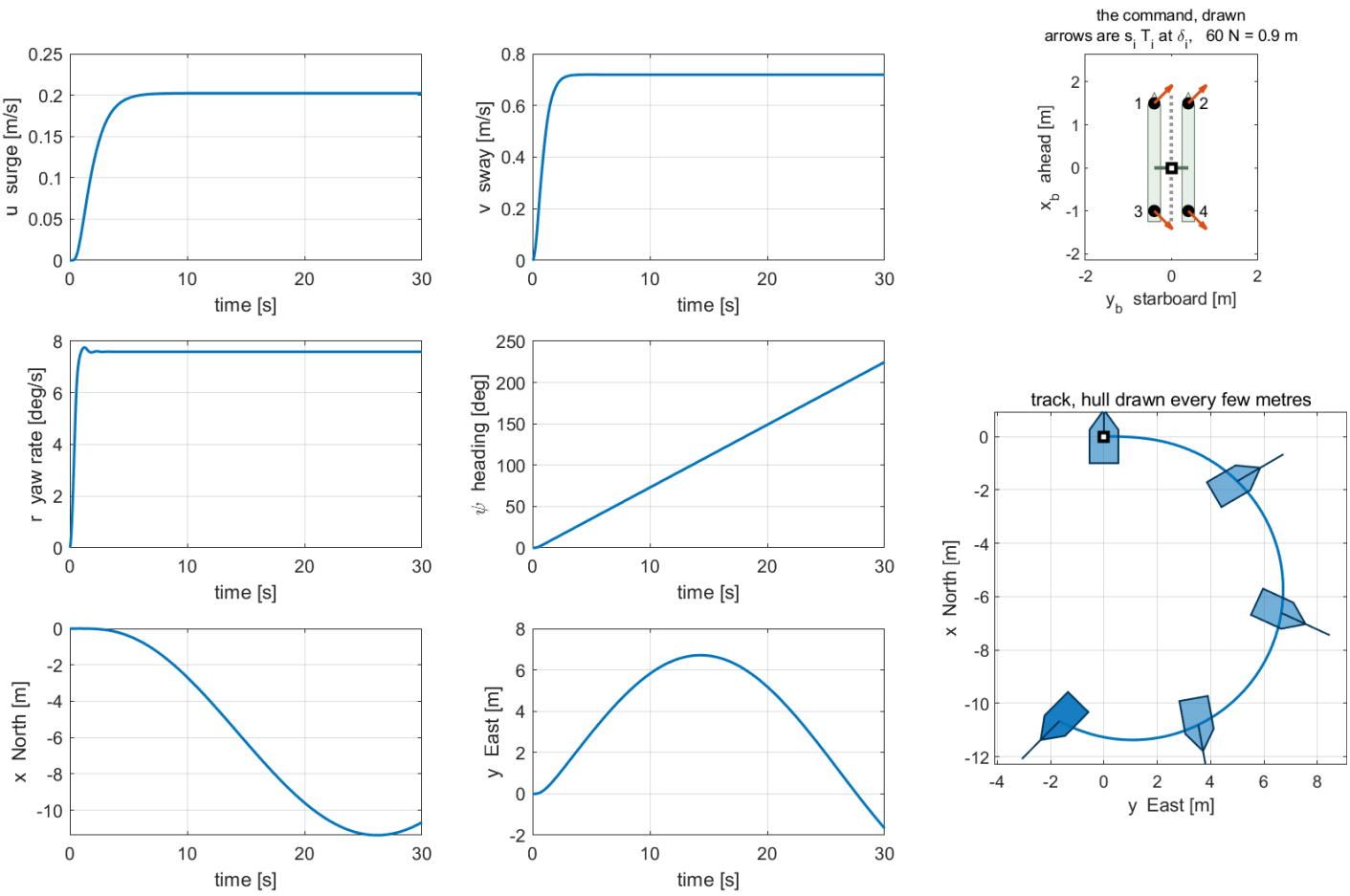




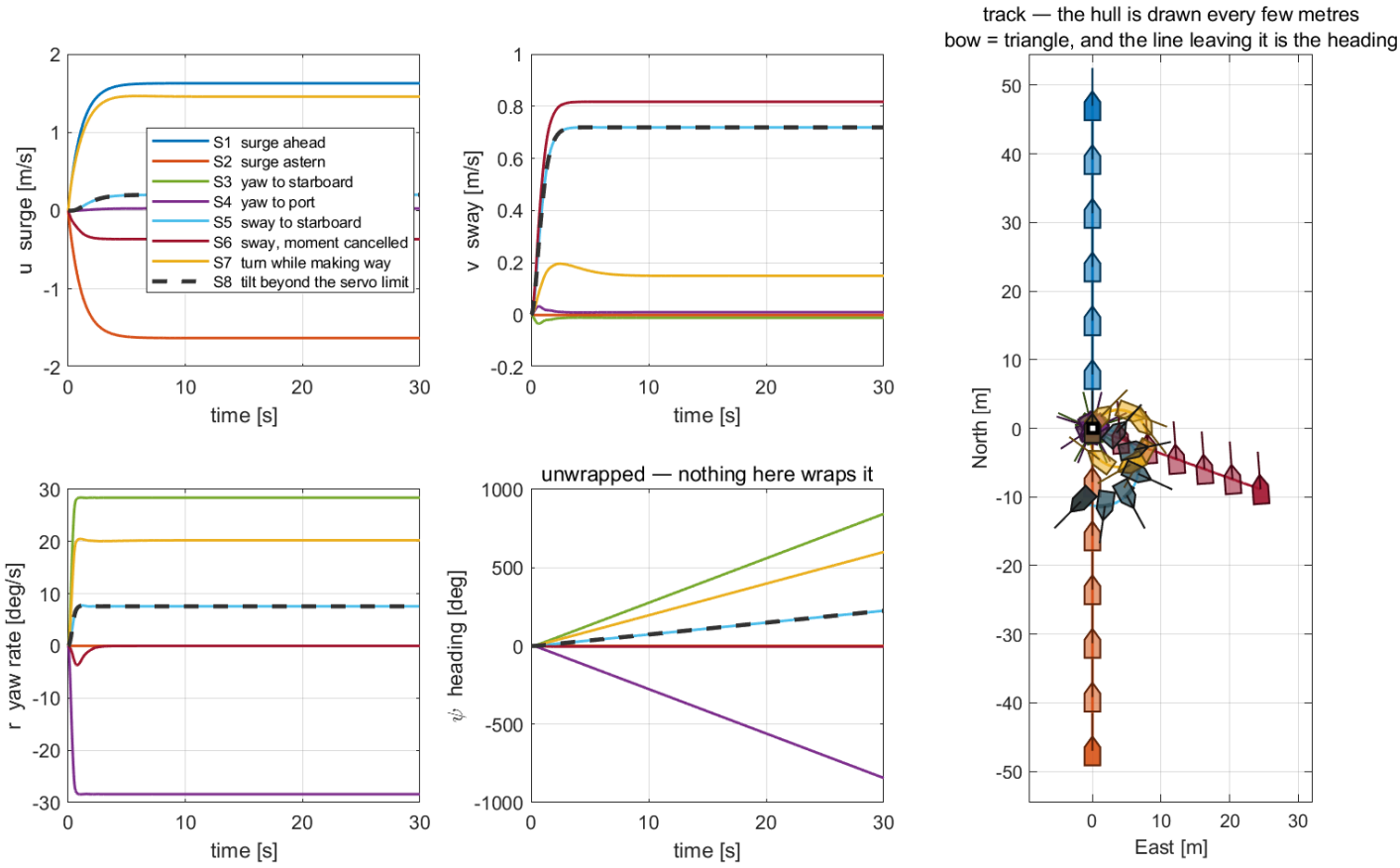




S8 tilt beyond the servo limit
n = [60.0 60.0 60.0 60.0] rad/s, δ = [70 70 -70 -70] deg, τ = [0.000 N, 112.820 N, 28.205 N m]
expected: identical to S5 in every number: the servo gave 45 deg, not 70



The overlay below is kept for comparison only. S8 is drawn dashed over S5, and the two curves coincide.



What to try	What to watch
`T_final = 120; animate = 1;` then Run	the hull drawn along the circle; S5 closes its 5.64 m circle in about 47 s
`W12_V_openloop({'S5','S6'})`	two sheets and no overlay file written — a partial run never overwrites the figure this section embeds

> ✎ **In class**
>
> - **Purpose** — connect a force in body axes to a track on the water, with nothing in between.
> - **Point to** — the S5 hull symbols on the track: the bow points well off the direction of travel.
> - **Ask** — "S7's force is at $28.74°$. Why does the hull crab at only $5.89°$?" Because sway is damped far harder than surge; the equilibrium is not the force direction.
> - **Take away** — an open-loop command gives a repeatable circle, and a circle is not a mission.

12-6. Whether the model is the equations

This section answers: table (1) showed the equations are right — what shows that the Simulink model evaluates those equations?

What is observed. Experiment 12-6 re-integrates each of the eight 30-s runs outside Simulink, with the same fixed-step `ode4` and the same step, driven at every step by the (n, δ) the model actually applied. The worst disagreement in position or speed over all eight runs is exactly $0.00\text{e}{+00}$.

- the driving signal is the **post-servo** pair read back from the log, not the command
- the heading column disagrees at 1.02×10^{-13} deg at worst, which is the unwrapping, not the dynamics

The result. Two different checks, and neither substitutes for the other.

Check	What it would catch	What it cannot catch
table (1) — hand formula against `otter_tilt4`	a wrong column, a wrong sign, a wrong arm	a model wired to the wrong signal, a wrong solver, a wrong step
table (2) — the model against independent re-integration	wiring, solver, step, saturation order	an error present in `otter_tilt4` itself, which both sides share

Derivation, one line at a time.

1. Table (1) compares two evaluations of the same function at one state. It therefore tests the function and nothing downstream of it.
2. A test of the model must re-run the model's own trajectory from an independent integrator. Writing the integrator by hand, in four lines of Runge–Kutta, removes Simulink from the comparison entirely.
3. The independent run must be given the same input the model received, which is **not** the command: the servo clipped it. Reading (n, δ) back out of the log, where they are tapped after the saturation stage, supplies the input actually applied.
4. The same fixed-step `ode4` at $h = 0.02$ s must be used. A variable-step solver on one side and a fixed step on the other would disagree at the step-error level and say nothing about the wiring.
5. A single-instant acceleration comparison would not do this job. The log carries 6 of the 12 states — $[u\ v\ r\ N\ E\ \psi]$ — and under sway the missing six, among them w, p, q and ϕ, are not zero. An acceleration reconstructed from the logged six would be evaluated at a state the vessel was never in.
6. Re-integration avoids that trap because it never reads the model's state: it starts from the same $\mathbf{x}_0$, which is known in full, and advances its own twelve states.
7. Agreement to $0.00\text{e}{+00}$ in position therefore says the two integrations performed bit-identical arithmetic, which is the strongest statement available: same equations, same inputs, same solver, same step.
8. The heading residual of 1.02×10^{-13} deg is not a dynamics disagreement. The log stores ψ in degrees and the comparison takes the smallest signed angle, so the residual is the cost of two conversions.

Experiment 12-6 · The model against an integrator written by hand (20 min)

What it measures. Lines 3, 4 and 7: that the wiring, the solver and the saturation order are what the equations assume.

The model. The same eight runs. The comparison lives in `W12_tab_reint`, which calls `otter_tilt4` directly inside its own Runge–Kutta loop.

Opening and running.

```
W12_V_openloop                % 블록 (2) 가 재적분을 한다 / block (2) re-integrates
```

Expected output, the second table:

```
(2) the Simulink model against the same equations, integrated outside it

    the same fixed-step ode4 and the same step 0.020 s, started from x0,
    driven at every step by the (n, delta) the model actually applied —
    the post-servo values read back from the log

  case   max |dN| [m]  max |dE| [m]  max |dpsi| [deg]  max |du| [m/s]   clip
  ------------------------------------------------------------------------------
  S1        0.00e+00      0.00e+00          0.00e+00        0.00e+00   n 0  d 0
  S2        0.00e+00      0.00e+00          0.00e+00        0.00e+00   n 0  d 0
  S3        0.00e+00      0.00e+00          1.02e-13        0.00e+00   n 0  d 1
  S4        0.00e+00      0.00e+00          1.02e-13        0.00e+00   n 0  d 1
  S5        0.00e+00      0.00e+00          2.54e-14        0.00e+00   n 0  d 1
  S6        0.00e+00      0.00e+00          0.00e+00        0.00e+00   n 0  d 1
  S7        0.00e+00      0.00e+00          1.02e-13        0.00e+00   n 0  d 1
  S8        0.00e+00      0.00e+00          2.54e-14        0.00e+00   n 0  d 1
  ------------------------------------------------------------------------------
  worst 0.0e+00 m or m/s over the 8 30 s runs
```

Reading the table against the derivation.

Where to look	What is there	Which line predicts it
`max dN` , `max dE` , `max du`	exactly zero, all eight runs	line 7
`max dpsi`	at most 1.02×10^{-13} deg	line 8 — the conversion, not the dynamics
`d 1` on S3 to S8	the tilt saturation bound during those runs	line 3 — the input re-integrated is the clipped one
`n 0` everywhere	the shaft limit never bound	$60 < 103.931$ and $73.485 < 103.931$

What to try	What to watch
`h = 0.05;` then `W12_V_openloop`	still zero. The agreement is between two integrators, so a coarser step moves both together
drive the re-integration with `n_cmd` instead of the logged `n`	S8 stops agreeing, because the command is $70°$ and the plant received $45°$ — line 3

✎ In class

- **Purpose** — separate a check of the equations from a check of the model, and show that this week has both.
- **Point to** — the `clip` column. The re-integration agrees **because** it was given the clipped input.
- **Ask** — "Why not just compare accelerations at one instant?" Because the log carries six of twelve states, and the other six are not zero under sway.
- **Take away** — zero is a stronger result than small, and it is only available when both sides run the same solver on the same input.

Part 2 · Laboratory run order

The experiments of Part 1 are worked through in order; this table is the index of what was run, for repeating the week at home.

Experiment	Model	Script	What it shows
12-0	`W12_openloop`	`W12_1_build_openloop`, `W12_0_setup`	the model written from code, and every constant
12-1	none	`verify_otter_tilt4`	four identities; $\delta = 0$ returns `otter.m`
12-3	`W12_openloop`	`W12_V_openloop` → `W12_tab_formula`	table (1) — force against the column rule
12-4	`W12_openloop`	`W12_V_openloop` → `W12_tab_track`	the three asymmetries, each against a row
12-5	`W12_openloop`	`W12_V_openloop` → `W12_plot_scenario`	the eight sheets `img/W12_result_S1.png` … `_S8.png`
12-6	`W12_openloop`	`W12_V_openloop` → `W12_tab_reint`	table (2) — the model against hand re-integration

Selecting runs: `W12_V_openloop` for all eight, `W12_V_openloop('S5')` for one, `W12_V_openloop({'S5','S6'})` for a pair, `W12_V_openloop(1:4)` by number. A partial run writes only the sheets it ran and leaves the overlay file alone.

> **One run order, four sections**
>
> Sections 12-3 to 12-6 share the **same eight 30-s runs**: table (1) measures the force at the start of each, table (2) re-integrates the whole of each, table (3) reads the end of each, and `W12_plot_scenario` draws each. Splitting them would send the vessel out thirty-two times for eight trajectories, and the tables would then be read against runs only nominally the same. The runs happen once, in `W12_V_openloop`, and each section's block is a file of its own — `W12_cases` fixes the eight commands, `W12_tab_formula` is §12-3, `W12_tab_track` is §12-4 and §12-5, `W12_tab_reint` is §12-6. The arrow names the file to open for that section.

Summary

Week Summary

Step	What was done	How it was verified
1	B built for four tilting units from the Week 3 column rule, with the mounting signs	`verify_otter_tilt4` prints 3×8, rank 3, null space 5
2	`otter.m` changed in four marked places into `otter_tilt4.m`	identities (a) and (b) exactly zero: $\delta = 0$ reproduces `otter.m`
3	the trigonometric form checked against Bf	identity (c), 7.19×10^{-14} over 200 random states
4	an open-loop model built, eight hand constants to the plant	`W12_1_build_openloop` prints overlaps 0, diagonals 0
5	the force of eight commands measured out of the plant	table (1), worst residual 2.8×10^{-14} N or N·m
6	the mirrored mounting shown to make surge symmetric	S1 and S2 both reach $\lvert X \rvert = 126.180$ N despite $k_{pos} \neq k_{neg}$
7	the residual yaw of a symmetric sway derived	$N = \bar{x}Y = 28.205$ N·m, matching S5 and S8 exactly
8	the thrust ratio that removes that yaw derived and run	$T_a = 1.5T_f$, i.e. $n_a = 73.485$ rad/s: $N = 0.000$ N·m at $X = -28.205$ N
9	the servo limit shown to act inside the plant	S8 asks $70°$, table (1) records $45°$, and S8 lies on S5
10	the eight tracks read at $t = 30$ s, one sheet each	table (3) and `img/W12_result_S1.png` … `_S8.png`
11	the model checked against an independent integrator	table (2), 0.00e+00 m and m/s over eight 30-s runs

Progress Check

> ★ **Minimum condition for following the next week**

Theory

- [ ] Able to state the column rule and apply it to one tilting unit, obtaining both of its columns
- [ ] Able to say why B is 3×8 and what the five null-space dimensions mean for an allocator
- [ ] Able to say where the mounting sign s_i sits in B, and why the two pairs have different zeros but the same sense
- [ ] Able to name the four places `otter.m` changes, and say why the damping is not one of them
- [ ] Able to derive $N = \bar{x}Y$ with $\bar{x} = (l_f - l_a)/2$, and predict 28.205 N·m without the script
- [ ] Able to say why re-integrating the logged input is a different check from table (1), and why a single-instant acceleration comparison is neither

Laboratory

- [ ] `verify_otter_tilt4` ran and printed `PASS` with identities (a), (b) and (d) exactly zero
- [ ] `W12_1_build_openloop` ran and printed overlaps 0, diagonals 0
- [ ] `W12_V_openloop` ran and wrote all three tables and the eight sheets
- [ ] `W12_V_openloop('S5')` ran alone and rewrote only that one sheet
- [ ] A ninth case was commanded by hand from `W12_0_setup`, and its X, Y and N were predicted before the run

Recorded observations

- [ ] The residual yaw measured for a symmetric $45°$ sway command, and the arm it comes from
- [ ] The crab angle of S7, and why it is not the $28.74°$ of its own force

---

Assignment 12

- **Due**: before the Week 13 session
- **Submit**: code, results, and a short analysis

① Requirements

Three runs, each one eight numbers given by hand in `W12_0_setup`. No allocator, no controller.

1. **Reproduce the yaw-free crab.** Apply §12-4 line 8 to obtain the stern shaft speed that cancels the residual yaw of a $45°$ four-unit sway command. Derive it rather than copying 73.485 rad/s: show the step from $\sum x_i T_i = 0$ to $T_a = 1.5 T_f$ and the conversion to shaft speed, including which thrust coefficient applies and why. Report the surge force the cancellation costs.
2. **Find a motion that is neither.** Choose eight numbers for which $X = 0$ **and** $N = 0$ while $Y \neq 0$ — a pure sway, with no surge and no yaw. Keep $\delta_i \in \{+45°, -45°\}$ so that only the four shaft speeds are free. State the two linear conditions on $T_1 \dots T_4$, solve them by hand, convert each T_i to a shaft speed with the correct sign-dependent coefficient, and check every value against the shaft limits before running.
3. **Break it deliberately.** Take the solution of requirement 2 and change one shaft speed by 10 %. Report what appears in X, Y and N, and say which of the three asymmetries the change exposed.

② Verification — mandatory

> **★ A claim is not a result. Numbers are required, with the averaging window stated**

What to report	How
hand-predicted X, Y, N for each of the three runs	from §12-1 line 7, before running anything
measured X, Y, N	table (1) of `W12_V_openloop`, or the same arithmetic in a script written for the purpose
the residual between the two	one number per run, and it must be below 10^{-12}
U, β and r at $t = 30$ s	read at $t = 30$ s, as table (3) does; state the instant, not an average
the clip flags	whether the shaft or the servo limit bound during the run

A run whose predicted and measured forces disagree by more than rounding is a mistake in the prediction, not in the model. Find it before reporting.

③ Analysis (5–10 lines)

Requirement 2 asked for a pure sway and the vessel can produce one, while requirement 1's command cannot. Explain which freedom made the difference, counting it against the five null-space dimensions of §12-1. Then state what an allocator would have to do that was not done by hand here, and why doing it by hand would not survive a changing demanded force.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

- Only faults that have actually occurred. Hypothetical entries are not included

Symptom	Cause	Fix
A tilt of $70°$ or $90°$ produces exactly the $45°$ result	the servo clip inside `otter_tilt4`, §12-2 change 3	expected. Read the `delta` column of table (1), which prints what the plant received
Four equal $\delta_i = +45°$ at four equal n_i produce no force at all, only a spot turn	the mounting signs: $s_3 = s_4 = -1$ reverses the stern pair's force, so the four cancel in surge and sway	expected, and it is row S3. A sway needs the two pairs at **opposite** δ, which is S5
A symmetric four-unit sway command yaws at 7.589 deg/s	$\bar{x} = 0.25$ m, §12-4 line 7	expected. Apply the ratio of §12-4 line 8 to cancel it, and accept the surge it costs
`W12_result_openloop.png` does not change after a run	a run of fewer than eight scenarios deliberately does not overwrite it, so the figure §12-5 embeds always carries all eight	run `W12_V_openloop` with no argument
A Constant block shows `-C-` instead of its value	`mss_style` sets Constant width to 55 px, and a longer expression does not fit	keep the Value expression short — the model uses `d_cmd`, not `delta_cmd`
`n_cmd = 600` saturates immediately	the command is rad/s, not rpm	60 rad/s is about 573 rpm; the limit is 103.931 rad/s
`W12_V_openloop` runs but no figure appears	the script was called before `W12_0_setup`	run `W12_0_setup` first; it defines `animate`, `pace` and `h`

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, §11.2 (thrust allocation and the configuration matrix), §7.3 (manoeuvring models and damping)
- MSS toolbox, `Tools/MSS/VESSELS/otter.m` — the hull, the propeller coefficients and the shaft limits used unchanged
- MSS toolbox, `demoOtterUSVPathFollowingHeadingControl` — the block-level conventions this course's models follow

In this course

- Week 3 — the column rule, and the only definition of $\boldsymbol{B}$ this course uses
- Week 5 — saturation, and what a limit does to a command
- Week 6 — `otter_bow.m` : changing the motion model rather than adding to its input
- `_tools/otter_tilt4.m` — the four changes, each marked `% ---- CHANGED (i of 4)`
- `_tools/verify_otter_tilt4.m` — the four identities of Experiment 12-1

Next Week

- **Week 13 — Allocation for Four Tilting Thrusters**
- The five null-space dimensions counted in §12-1 are spent: choosing eight numbers that produce a demanded $\boldsymbol{\tau}$, and choosing among the many that do
- What must be prepared: the three asymmetries of §12-4 stated from memory, with the numbers 0.5817, 0.25 m and $45°$ attached to the right one